Disability in the kinship network: population-level exposure to kin with disability across ages in Europe

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Abstract

Disability is usually framed as an individual condition, yet its associated consequences may extend through kinship networks. Despite this, the population reach of disability through kin remains unknown. We use demographic kinship matrix models combining fertility and mortality schedules with disability prevalence, to estimate population exposure to disability through grandparents, parents, siblings, children, and grandchildren across ages in Europe in 2021. We quantify exposure using three complementary measures: the probability of having at least one kin with disability, the population-level number of individuals exposed, and the proportion of kin with disability. Our results show that exposure is both widespread and substantial: across all ages and kin types, 70% of women in Europe, corresponding to about 160 million women, have at least one close kin with disability. In childhood and early adulthood, more than half to three-quarters of European women have at least one grandparent with disability, with the age-specific number of exposed women peaking at about 1.5 million; in midlife, about one-third to nearly one-half have at least one parent with disability, with the population-level exposure peaking at about 1.4 million women. Throughout much of women's lives, between 20% and 40% of their close kin have disability. These findings show that disability is not only an individual experience but also a relational exposure amplified by kinship structures. Exposure to kin disability constitutes a hidden source of inequality that accumulates over the life course and intersects with gender and socioeconomic inequalities.

Significance

The disadvantages associated with disability extend through kin networks, affecting kin lives. This study shifts the traditional perspective from considering disability as an isolated individual condition to considering disability as a relational feature embedded in kinship networks. This matters for understanding care demands, cumulative inequalities, and policies to support people with disability and their kin. To quantify exposure to kin with disability in ways not possible with traditional surveys or conventional individual-level data, we use demographic kinship models. Even when disability is experienced directly by only a share of the population, kinship structures multiply its reach: one person with disability may have relatives who experience disability indirectly through care, concern, emotional strain, time allocation, and changes in family life.

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Introduction

Disability is often treated as an individual characteristic, a lasting legacy of the medical model that has historically defined disability as a purely "medical impairment" (1). Yet people do not experience life as isolated individuals. They live within families and kinship networks, where one person's disability can alter the opportunities, responsibilities, health, and well-being of many others. The societal reach of disability therefore extends beyond those directly affected, but we still know remarkably little about how far it reaches, how many people it touches, or how this exposure changes across the life course.

With the introduction of the *International Classification of Functioning, Disability and Health* (ICF) in 2001, the World Health Organization (WHO) formally conceptualized disability as arising from the interaction between an individual's health conditions and the social environment in which they live (2). This perspective shifted disability from being understood solely as an individual impairment to being viewed as a socially embedded phenomenon. Yet one important dimension of this embeddedness has received comparatively little attention: kinship. Since birth, individuals are generally embedded in kinship networks (3, 4) that connect them to other individuals across the life course, such as parents, grandparents, siblings, children, and grandchildren. Kin networks constitute a crucial social structure that shapes individuals' opportunities, resources, and constraints (4, 5). Kin can provide companionship, practical assistance, and long-term support, often reducing social barriers, constituting a latent web of support and a relational reserve that is activated in times of need to safeguard its members throughout their life courses, becoming the most important resource when disability arises (6–9). In this sense, whether individuals' kin members are providers or recipients of care is strongly driven by their health status. Through these networks, the disadvantages and negative consequences that people with disabilities face in their daily lives extend well beyond the individuals who directly experience disability. Disability may, in fact, involve needs for care, support, and adaptation that may have consequences on multiple kin simultaneously. Understanding this broader reach is increasingly important in ageing societies, where disability may shape not only individual well-being but also family care demands, intergenerational obligations, and the accumulation of social and economic disadvantages. However, these patterns remain poorly understood because most available data sources usually capture kin links and health only within households, for selected relationships, or in specific national contexts, leaving exposure to disability across the broader kin network (also including non-coresident kin) largely unmeasured. Disability itself is far from rare: worldwide, more than 1 billion people (around 16% of the world population) have a disability (10), and around 90-100 million people in Europe (11). Despite this, its broader relational reach remains largely unmeasured. In fact,

we still lack systematic and population-level evidence on how many people are exposed to disability of a kin and how such exposure is distributed across ages and kin types.

This gap of knowledge is particularly striking because kin are central to how care and support are organised in most societies. Close family members are often the primary providers of long-term assistance (7), representing not only a source of social capital and reserves but also the potential pool of informal caregivers on which individuals and welfare systems rely. Having a kin member with disability can therefore shape individuals' lives in important ways, including: adverse health outcomes such as increased psychological distress, physical strain, and even elevated mortality risks (12, 13); changes in family trajectories and household organisation, influencing fertility expectations, partnership dynamics, and the timing of family formation when people anticipate or assume caregiving roles (14, 15); reduce labour market participation through lower working hours, transitions to part-time work, or labour-force exit, with long-term implications for income, pension accumulation, and economic security, as indirect costs (16–20); direct costs related to treatment, rehabilitation, assistive technologies, transport, or the purchase of formal support when public provision is limited (17, 21). These consequences often accumulate over the life course and intersect with broader axes of stratification, and especially with gender inequality, as women continue to assume a disproportionate share of care-related family responsibilities (22, 23). At the same time, exposure to disability within families should not be understood only through a "burden" lens: having a kin with disability may also foster solidarity, empathy, and prosocial orientations within kin relationships (24–27). Although much has been learned about the consequences of disability for kin members and its associations with individual-level outcomes, a basic first step is to establish how many people are exposed to disability through kin, and how this exposure varies by age, gender and kin type. This makes measuring exposure to disability in kin networks crucial, not only for identifying potential caregiving demands, but for revealing how widely disability may shape the lives and outcomes of people beyond those directly experiencing it.

In line with the ICF framework developed by the WHO (2), we rely on a definition of disability as a multidimensional outcome arising from the interaction between health conditions, functional limitations, and the social and physical environments in which people live, affecting individuals' ability to carry out everyday activities and to participate fully in social life, rather than as a fixed attribute of the individual alone. Despite disability being usually framed only at the level of the individual, its implications are inherently relational. Kinship creates, in fact, a multiplier effect: when one person experiences disability, multiple kin may be affected at the same time through care responsibilities, financial consequences, altered family roles, and the need to navigate institutions and environments that are more or less inclusive. The presence of disability in one generation may

also reverberate across others simultaneously: middle-aged adults may provide support to ageing parents while still caring for children (28), and the health status of grandparents may shape their ability to remain involved in intergenerational support.

Moreover, over the life course, individuals are themselves increasingly likely to experience disability, as reflected in age-specific prevalence and risks. Disability is therefore not confined to a small subgroup of the population. Rather, in different ways, it is encountered by many (and likely most) individuals either directly, indirectly through their kin, or both. This perspective shifts attention from disability as an isolated or marginal condition to disability as a widespread and interconnected feature of social life. For this reason, it also reinforces the urgency of creating more inclusive societies and environments. If disability is a widespread and interconnected feature of social life, then non-inclusive institutions, inaccessible spaces, discriminatory norms, and the treatment of disability as something "isolated" or "different", have consequences that extend beyond those directly experiencing it: they also shape the lives of kin, amplifying disadvantage across family networks and across generations.

Understanding exposure to kin disability is especially important in the context of demographic change. Across Europe, long-term declines in fertility and mortality have profoundly reshaped kinship structures (29, 30). Historically, higher fertility and lower survival produced kinship systems with broad horizontal ties and relatively short vertical overlap. As fertility declined and longevity increased, kinship networks shifted toward more "beanpole" structures: fewer members within each generation, but longer vertical extension across generations (29). In these thinner yet longer families, horizontal kin such as siblings become less numerous, while vertical ties such as parents, grandparents, children, and grandchildren are more likely to coexist for longer periods. At the same time, postponement of childbearing has widened age gaps between generations and shifted overlap toward older ages. Together, these changes mean that kin networks are ageing. Individuals may now spend more years coexisting with older parents and grandparents at ages when disability becomes more prevalent, while the pool of kin available to share responsibilities may shrink. On the one hand, smaller families may reduce some forms of multiplication of disability exposure simply because there are fewer kin. On the other hand, older kin networks imply more older relatives, for more years, at ages when disability risks are higher. Future disability exposure may therefore become concentrated within older and numerically thinner kinship networks, potentially intensifying pressures (in terms of care demands) on fewer individuals.

European populations have undergone fertility decline, mortality improvement, and epidemiological change at different times and with different intensity, producing substantial variation in kin availability, kin composition, and the age structure of kin networks (29–31). Southern

European countries, characterised by very low fertility and rapid population ageing, tend to exhibit vertically extended but numerically smaller kin networks, while Northern and some Western European countries often retain somewhat larger kin pools because of higher fertility combined with similar low mortality. Central and Eastern Europe followed yet other demographic trajectories, marked in many cases by persistently lower life expectancy and high avoidable mortality (32), yielding more heterogeneous patterns of kin availability and characteristics. European countries also differ in the extent of investment in prevention and healthy ageing (33), which may contribute to cross-national variation in the timing and pace of epidemiological transition, and thus in the resulting health and disability profiles. In addition, countries differ in how care responsibilities are organised between families, the state, and the market (30, 34–37). This means that, not only the magnitude, but also the implications of exposure to kin disability may vary across contexts in Europe: in more familistic settings, such exposure is more likely to translate into direct and heavier family caregiving responsibilities, whereas in more de-familialised settings formal services and income support may partly mitigate these pressures.

Understanding patterns of exposure to kin with disability is usually difficult because data rarely capture whole kinship networks directly, together with their health status. In most surveys, administrative sources, and population registers, detailed kin links are observed only within households, for selected relationships, or for specific national contexts, leaving the broader family network and its health status largely untracked. Formal demographic kinship models have become a crucial tool for estimating kin availability and kin structure from fertility and mortality schedules. A growing literature has used these models to show how demographic regimes shape the number and composition of kin across ages and populations (3, 30, 39–45). Some recent work has also started to incorporate health conditions into kinship analyses. In particular, Feng and colleagues examined kin-based dementia exposure in the United States and projected the future dementia burden in China, showing how population ageing and family change redistribute potential care demands across generations (46, 47). These studies demonstrate the value of integrating the health component into the study of kinship, but evidence on disability exposure in the European context remains lacking. This leaves an important gap. Compared with specific late-life conditions, overall disability is broader and more prevalent, spans a wider part of the life course, and is likely to affect kinship networks in ways that are both quantitatively substantial and socially consequential. Yet we still don't know how many people are exposed to disability within kin networks, how this exposure changes with age, and from which kin relations.

Addressing this gap is important for at least three reasons. First, exposure to disability through kinship is likely to be widespread. Even when disability is experienced by a population

share at particular ages, most individuals are connected to several kin, so that one case may have consequences on multiple lives simultaneously. Second, this exposure shapes individual outcomes, such as health, labour supply, family trajectories, and economic security, and those effects may accumulate across the life course and reinforce existing inequalities. Third, a kinship perspective adds to existing arguments for inclusive environments by showing that the disadvantages associated with disability are not confined to individuals with disability, but also have consequences on the many relatives whose lives are organised around support, care, and adaptation. Viewing disability only at the level of the individual, therefore risks understating both its prevalence in everyday social life and its broader implications for inequality, intergenerational dependency, and welfare-state sustainability.

In this study, we quantify exposure to disability within kinship networks for women in Europe using demographic kinship models. We combine fertility and mortality rates with disability prevalence to reconstruct kinship networks and characterise their disability composition across individuals' ages. We focus on a restricted pool of close kin, namely grandparents, parents, siblings, children, and grandchildren, because these relationships are especially relevant for support and care (8, 48, 49). This pool includes the close kin of first-degree (parents and children) and second-degree (grandparents, grandchildren, and siblings), i.e. ego's ascending and descending lines, plus siblings, while excluding more distant kin and (because of data and/or methods constraints) focusing on consanguineal kin only. It allows us to follow four family generations, which have become increasingly important as vertical kinship relations lengthen. We present women as the main analytical case because the implications of kin disability exposure are strongly gendered, with women disproportionately involved in family caregiving. We assess exposure to disability in the kinship network through three complementary metrics: the probability of having at least one kin with disability, the absolute number of individuals exposed, and the proportion of kin with disability. Together, these measures capture how likely exposure is, how many people it reaches, and how large a share of kin it represents. By combining these perspectives, we conceptualise exposure to kin disability as a structural characteristic of kinship networks.

To do so, our analytical approach proceeds in four steps: (i) we draw on age-, gender-(, and country-) specific fertility and mortality schedules and the sex ratio at birth from 1950 to 2021 from the United Nations World Population Prospects (UNWPP), to reconstruct kinship networks in Europe, and to estimate the expected number of kin by kin type, gender, and age of a focal woman; (ii) we combine these kinship estimates with age, gender, and disability prevalence from EU-SILC survey to specify kin disability status (with or without disability); (iii) from these quantities, we derive the two complementary measures of exposure to disability through kin, namely the probability of

having at least one kin with disability and the proportion of kin with disability; and (iv) we use age-, gender-specific population counts from the UNWPP to translate the probabilities into population-level figures.

Figure 1 illustrates the logic of the three measures used in the study with a schematic example population of four (focal) women at same age (e.g. aged 35), each with five living kin, but with different kin networks: A has two parents, one sibling, and two children, with one parent having disability; B has one grandparent, two parents, one sibling, and one child, with no kin having disability; C has two parents, one sibling, and two children, with one parent and one child having disability; and D has one grandparent, one parent, two siblings, and one child, with the grandparent having disability. First, the probability measures the share of individuals who have at least one kin with disability at age 35; in the example, 3 of 4 women aged 35 have at least one kin with disability, yielding a probability of 0.75. Second, this probability can be translated into an absolute population-level estimate of individuals having at least one kin with disability, by multiplying the probability by the number of women at that age in the overall population of women aged 35 in the specific context (Europe in this case). Third, the proportion of kin with disability when women are 35 years old measures, in relative terms, how many of their kin have a disability: for the focal individual C in the example, 2 of 5 alive kin have disability, corresponding to the 40%. For the same individual, this 40% can be broken down into a 20% that is the share of kin who are parents and are with disability, and 20% that are children who are with disability, summing to a total of 40% share of overall kin with disability. This figure is purely schematic and is intended only to illustrate the logic of the measures. The circles, in fact, represent simplified examples of kinship networks in which some kin have disability, and others do not, but the empirical estimates in our analysis are not obtained by directly counting observed individuals in observed kinship networks (as these informations are often not available). Instead, for each focal age, gender (and country, or Europe), the kinship model uses fertility and mortality schedules to estimate an average kinship network with the expected number of living kin, and then combines these estimates with age- and gender-specific disability prevalence to derive the expected number of kin with disability. The measures of interest are then derived from these expected values.

Full details are provided in Materials and methods section.

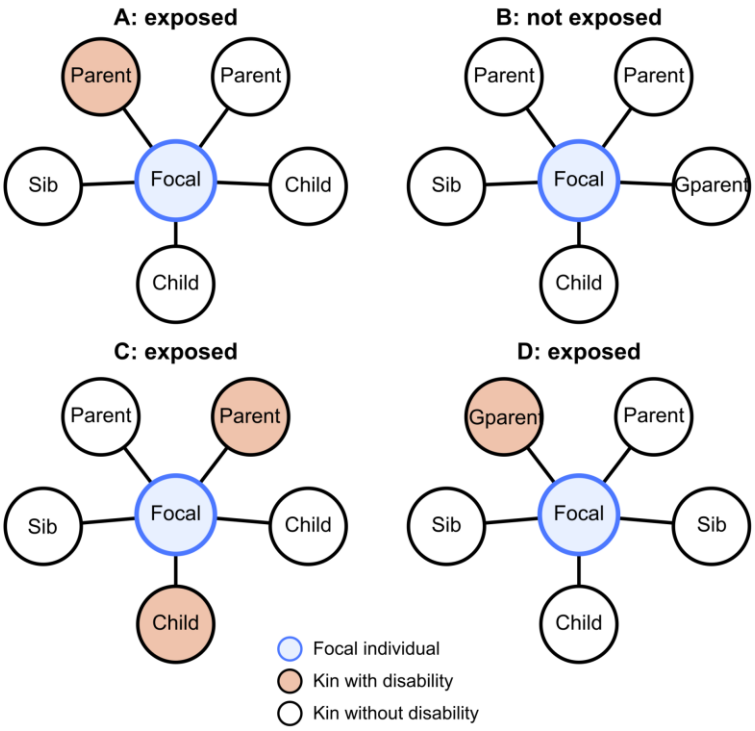


Figure 1: Illustrative example of some kinship networks, where kin are specified by type and disability status.

Results

Probability of having at least one kin with disability

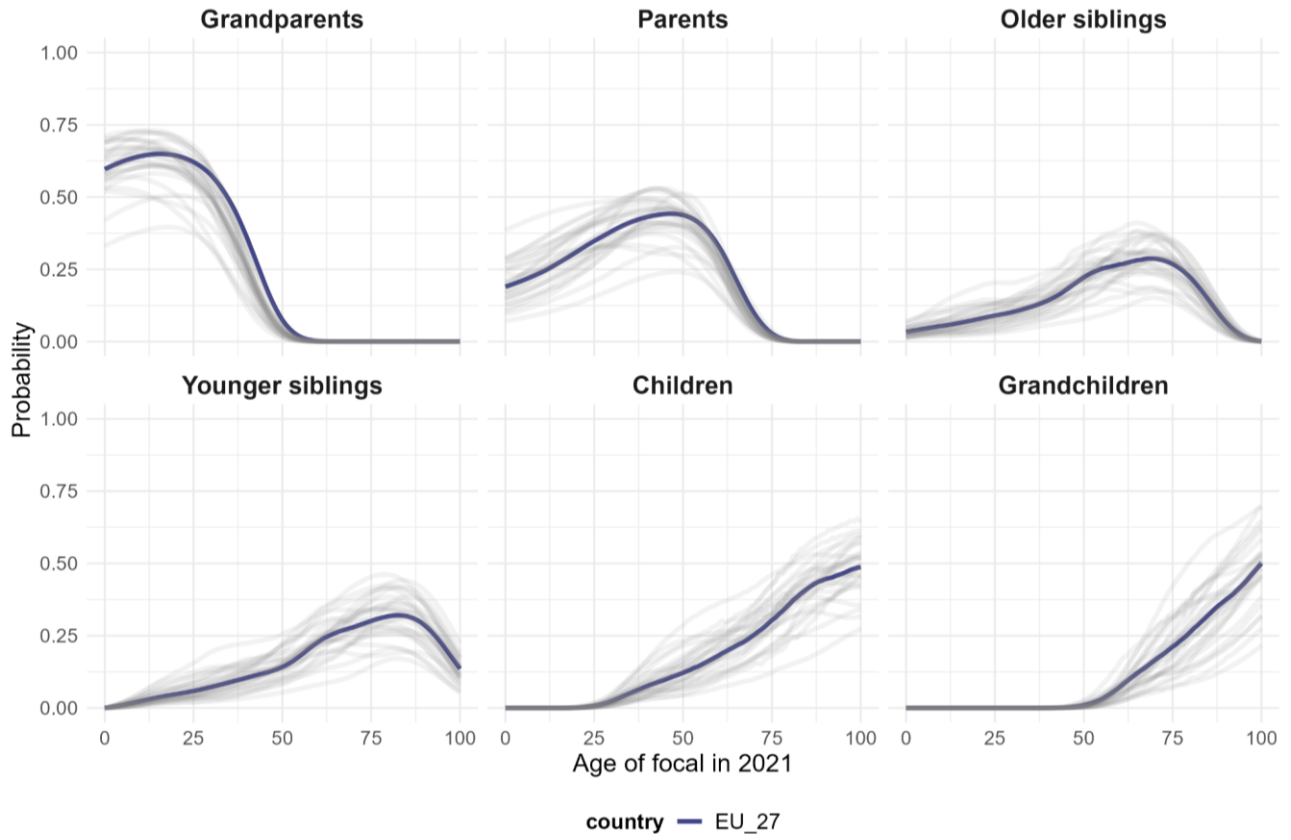
To understand how widespread exposure to disability through kin is across individuals, Figure 2, Panel A, shows the probability that women, at each age in 2021, have at least one kin member with disability, for each kin in Europe (EU-27). The probability of having at least one grandparent with disability is high in early life and rises sharply through childhood and adolescence, reaching its maximum in young adulthood, roughly between 50% and 75%, i.e. nearly two-thirds of individuals have at least one grandparent with disability at these ages, making grandparental disability a common form of exposure early in life. Thereafter, this probability declines relatively quickly across adulthood, approaching zero at older ages, reflecting low grandparental survival into focal's old age. The probability of having at least one parent with disability is already non-negligible in childhood and early adulthood and typically increases steadily later in life, as parents age. It peaks around age 50 (or shortly earlier), when individuals' probability of having at least one parent with disability reaches approximately 30 to 50% (i.e. around one third to half of individuals in midlife have a parent with disability), and then declines at women's older ages. In the case of siblings, the probability increases gradually through adulthood (as siblings age and disability becomes more

prevalent), and the probability reaches the highest levels in later life. At very older ages, the probability of having at least one sibling with disability is around or exceeds the 25%, indicating that exposure to sibling disability becomes more common in later life, exposing many older individuals who have siblings alive at older ages. In terms of exposure to disability among descendant kin, the probability of having at least one child with disability begins to rise in young adulthood as children are born and age, and continues increasing into mid and later life. By very old ages (85+), often more than 30% of individuals reaching those ages have at least one child with disability. The probability of having at least one grandchild with disability emerges later (as also grandchildren are born, from midlife onwards), and rises progressively at older ages of the focal individual as grandchildren accumulate and grow older. Importantly, the probability of having children and especially grandchildren with disability at very old ages remains relatively high as a result of both the selective survival of the focal individual (the focal individual being alive at very advanced ages) and the fact that the surviving descendants have themselves reached older ages, when disability becomes more prevalent. Country specific results are shown in the Appendix Figure S1.1 and S1.2.

The age-kin profiles in panel A reflect the changing composition of kinship networks over the life course. In fact, different generations overlap at different stages of the life course. Moreover, the likelihood that each individual, and thus each kin, experiences disability increases as they age. This means that the ageing and disappearance of older generations and the progressive emergence and ageing of descendants, together with the age-schedule of disability, shape which kin types contribute most to exposure at different ages. This exposure may also be cumulative, as individuals can encounter disability among different kin at different ages.

Figure 2, Panel B shows the probability (%) of having a kin with disability over all ages and kin in European countries. On average, in Europe there is a probability of around 70% of having at least a kin with disability, spanning from around 60 to 80% in most European countries (for instance, the likelihood of having a close relative with disability is low at around the 60% in Norway and particularly high at around 77% in Portugal).

A


Grandparents
Parents
Older siblings
Younger siblings
Children
Grandchildren
Probability
1.00
0.75
0.50
0.25
0.00
0
25
50
75
100
Age of focal in 2021
country
EU_27

B

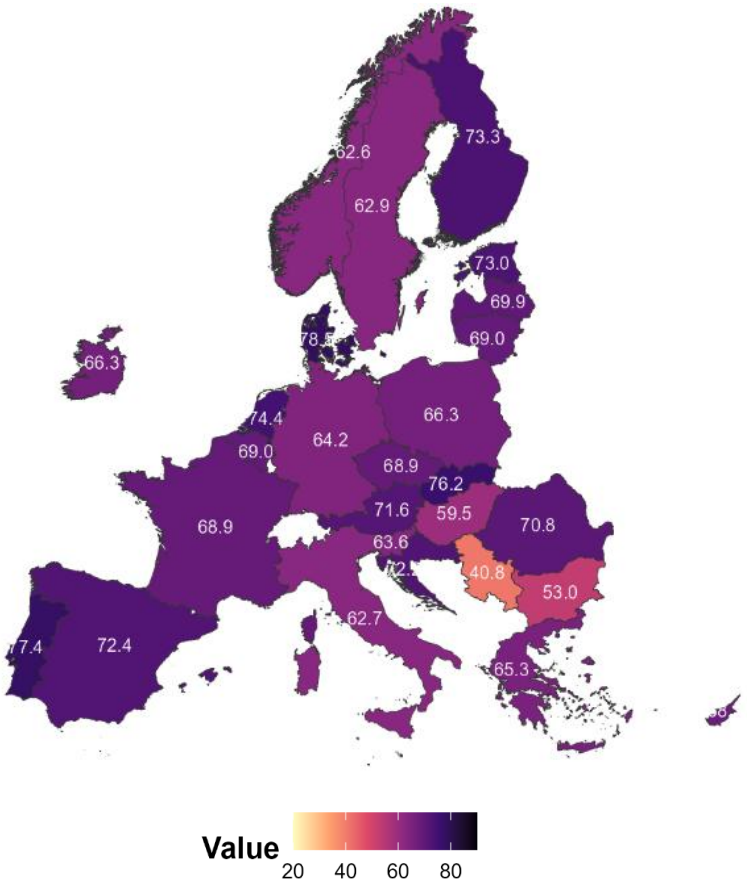


Figure 2: Probability (%) of having at least a kin with disability. Panel A: the probability is shown by kin type (e.g. probability of having at least one grandparent with disability, etc.) and age of a focal woman in 2021 in European countries, with the Europe (EU_27) value highlighted; Panel B: the probability is shown over all kin and ages in each European country.

Absolute number of kin with disability

Figure 3 translates the probability estimates into the absolute number of women who have at least one kin with disability by focal age and kin type in 2021, for the aggregate Europe (EU-27). Expressing the results in absolute numbers highlights the population-level magnitude of this exposure and shows how many individuals are likely to experience disability within their kinship networks at different stages of life. In overall Europe, exposure reaches very large numbers across women's ages: grandparent disability affects up to around 1.5 million women by age in childhood and early adulthood, parental disability peaks at around 1.4 million women by age in midlife, exposure through older and younger siblings reaches roughly 0.8 and 0.75 million women by age

later in life, and exposure through children and grandchildren rises at very old ages to around 0.7 and 0.5 million women by age, respectively. Exposure in absolute numbers is high early in life because of grandparent disability, followed in midlife by parental disability, and later by disability among siblings, children, and eventually grandchildren. These age and kin profiles mirror the dynamics described for the probability measure and reflect the changing composition of kinship networks over the life course. Country specific results are shown in the Appendix Figure S2.

As shown in Appendix Figure S3, Appendix Table S1 and Table S2, over all kin, across most of adult ages, more than 2 million women in Europe have at least one kin with disability; over all kin and over all ages, the magnitude of women exposed to kin disability in 2021 in Europe reaches around 160 million women in total (corresponding to about 70% of the total EU female population, i.e. an overall probability of 70% as shown in Figure 2B); this also translates into a "kin disability multiplier" indicating that for every person with disability, around 1.6 women are exposed to having at least one kin with disability.

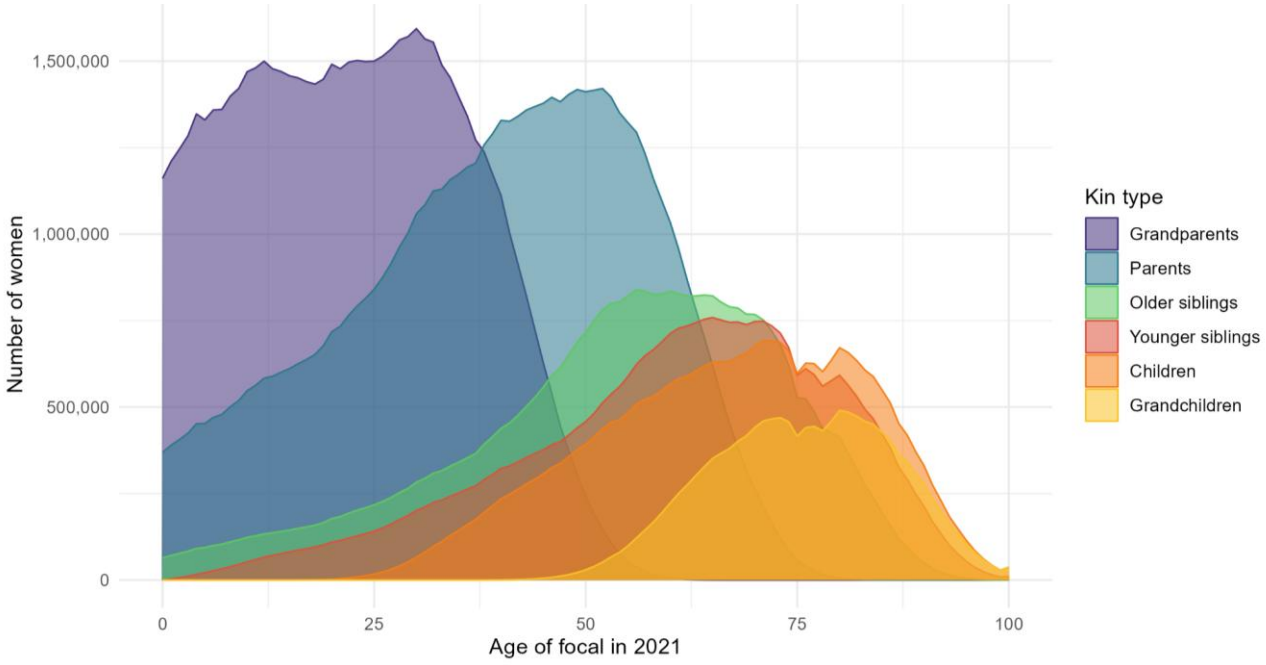


Figure 3: Absolute number of women having at least one kin with disability, by kin type and age of the focal woman in 2021 in Europe

Proportion of kin with disability

A complementary perspective is provided by the proportion of the total kin with disability, providing a measure of the magnitude or weight of such exposure, i.e., over the total available kin, how many are with disability. Figure 4, panel A shows the proportion of kin with disability across ages of a woman in Europe. The proportion of kin with disability increases during women's early life (childhood and youth), reaching its overall maximum of almost 30% around age 25, then declines until around age 40. A second, smaller peak appears in late adulthood (around ages 50–55) reaching around 25%, followed by another decline up to around age 60. After this point, the proportion tends to rise again (reflecting the ageing and accumulation of disability at older ages among kin).

These dynamics can be further understood if this overall proportion is broken down by kin type, as in Figure 4 panel C. The total proportion of kin with disability is first computed across all kin (as shown in Figure 4 panel A) and then subdivided by kin type, so that the stacked areas sum to the overall proportion, and each component represents the share of all kin that are of a given type and have a disability. In the first part of life (up to around ages 40–50), the proportion is mainly driven by the disability of ascendant kin. The first peak around age 25 largely reflects the disability in grandparents; the second peak around late adulthood is mainly due to disability in parents; at older and very older ages, exposure is increasingly driven by siblings, as well as by children and eventually grandchildren. Generally, and similarly to what has been discussed for the probability measure, these distributions reflect the changing kinship composition across the ages of individuals. In early life, kin are mainly composed of grandparents, parents, and siblings. During young and mid-adulthood (around ages 20–40), children start appearing and accounting for an increasing share of the kinship network. In later adulthood and at older ages (50+), grandchildren gradually appear, while older relatives begin to disappear, so that family composition shifts from being dominated by ascendants to being increasingly dominated by descendants.

Figure 4 panel B shows the proportion of kin with disability for a woman at age 25 in 2021, across European countries. This proportion typically ranges from around 15% to around 35% and some geographical patterns can be observed: countries in Northern and Western Europe often have lower proportions (Norway among the lowest), while countries in Southern and parts of Eastern Europe show higher proportions.

Figures for more countries and the map of the proportion at other ages are provided in the Appendix, Figures S4 and S5.1 to S5.4, respectively. Full kin counts (denominator) underlying the proportion estimates in selected countries, and the corresponding total counts of kin by disability status (numerator) are shown in Appendix Figure S6.1 and S6.2, respectively.

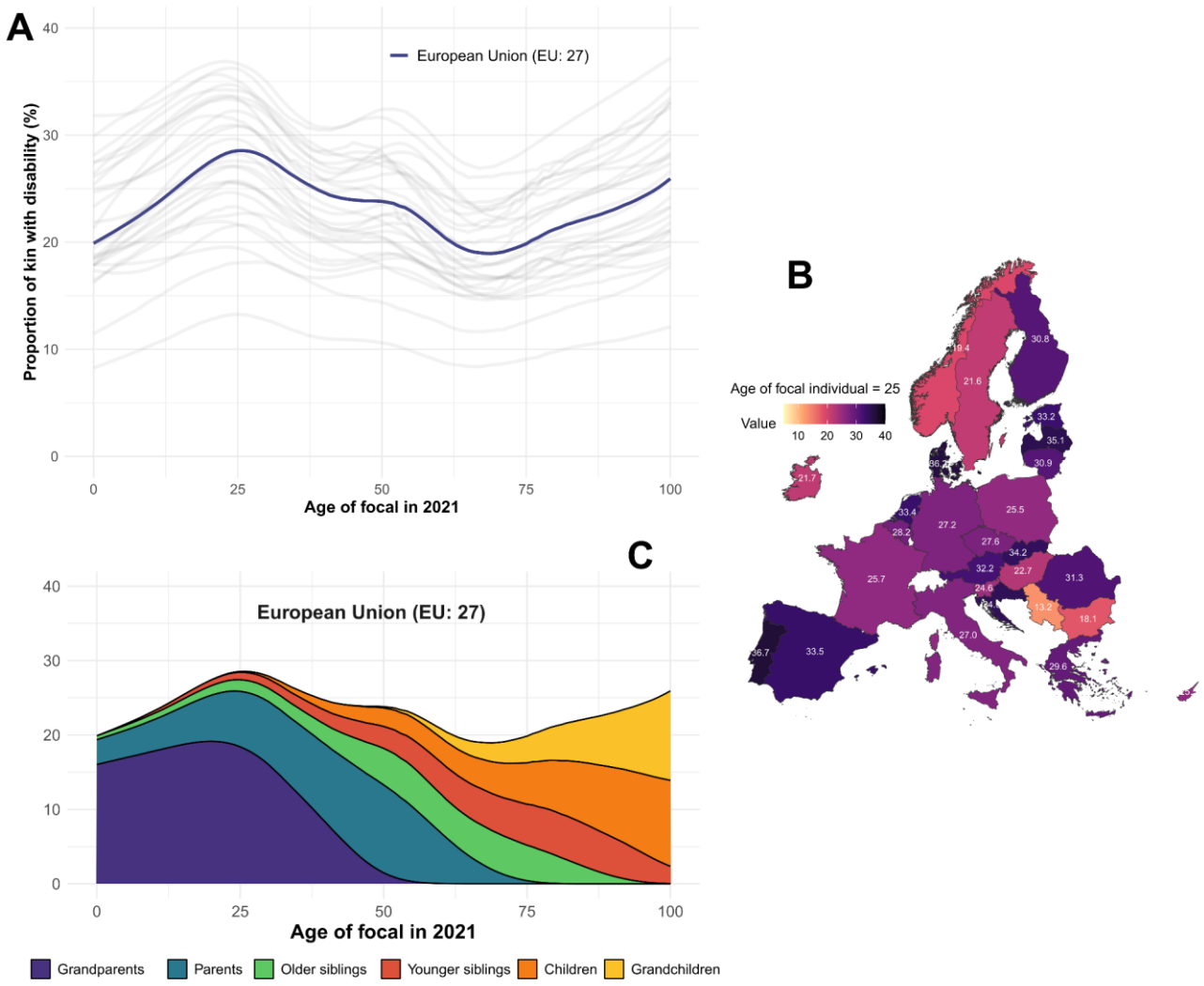


Figure 4: Proportion of kin with disability. Panel A: Proportion of kin with disability, across ages, for women in 2021 in European countries, with the Europe (EU_27) value highlighted; Panel B: Map of the proportion of kin with disability for 25-year-old women in 2021 in European Countries. Panel C: Specification of the proportion by kin type, across ages, for women in 2021 in Europe.

Robustness analysis

We conducted two sets of robustness checks to assess whether the main findings depend on modelling assumptions and to compare selected estimates with directly observed survey data.

First, because the probability of having at least one kin with disability is derived from expected kin counts, we examined whether the results are sensitive to alternative assumptions about the extent to which disability may be concentrated within families (see methods section).

Allowing for less or more clustering of disability in families slightly changes the absolute level of the estimated probabilities only modestly, shifting them upward or downward, while leaving the age profiles and cross-national differences essentially unchanged (Appendix Figure S7). The total range between the lower and upper probability bounds is generally around the 5% (about ± 2.5% around the main estimate). The same holds for the corresponding population-level estimates, which vary only within a limited range across alternative specifications (the lower-to-upper bound spans 5% of the main estimate, about ±2.5%), without altering the overall patterns or substantive conclusions (Appendix Figure S8).

Second, we compared selected model-based estimates with corresponding indicators computed directly from individual-level EU-SILC data, when possible. In this survey, kin links can be tracked only for coresident individuals. This comparison thus focuses on mothers, fathers, and children who live in the same household as the focal individual and which can be identified consistently in SILC data. Other kin types (such as grandparents) are less likely to be observed in the same household and therefore their estimates cannot be compared. Despite this more restrictive comparison, the model-based estimates are found to be broadly consistent with the household-based SILC ones, indicating that the model captures similar levels of exposure where direct validation is possible (Appendix Figure S9-12). We found some small differences, particularly for the probabilities at advanced age for parents and children, which tend to be slightly higher in the model-based estimates. This difference can be due to the empirical SILC measures that are restricted to coresident kin and therefore reflect a narrower subset of potential exposure, especially at later ages when kin are less likely to live together. Moreover, estimates from kinship models and surveys can also diverge at older ages because of small sample sizes in the latter (50). Previous research has shown that demographic kinship models closely reproduce observed kinship structures when compared with survey and population register data, supporting their validity in capturing overall kin availability (50, 51). Here we extend that validation to the health dimension of kinship networks by comparing model-based estimates of disability exposure with empirical measures derived from SILC data for coresident kin.

Overall, these checks show not only that the main findings are robust to alternative assumptions and broadly consistent with the survey evidence when direct comparison is possible, but also that the study contributes methodologically by validating the modelling approach in this setting.

Discussion

Mapping exposure to kin with disability is crucial for understanding how disability extends beyond the individuals directly experiencing it. Disability is not rare, and even less rare are the kin members whose lives are connected to it through care, support, and shared family responsibilities. By combining demographic kinship models with age- and gender-specific disability prevalence, this study makes it possible to quantify exposure to disability through close kin across ages and kin relations in Europe. Specifically, exposure to kin with disability in Europe is both widespread and substantial. In 2021, about 70% of the total EU female population, equal to around 160 million women, had at least one close kin with disability. The kinship networks produce a disability multiplier: for every person with disability, around 1.6 women are exposed to disability through at least one close kin member. Exposure also has considerable intensity. At the ages when kin disability is most concentrated, the proportion of close kin with disability can even reach about 35%. This exposure follows a consistent life-course pattern. In childhood and young adulthood, it is mainly driven by grandparents: around one-half to three-quarters of women have at least one grandparent with disability. In midlife, exposure shifts toward parents, with roughly one-third to one-half of women having at least one parent with disability. At older ages, exposure through siblings and descendants becomes increasingly common, as these kin age and disability risks rise. When translated into population counts, these patterns involve very large numbers of women: up to about 1.5 million women at each age are exposed through grandparents in childhood and early adulthood, and about 1.4 million through parents at each age around midlife. Taken together, these findings show that disability is not confined to a subgroup of individuals, but constitutes a high and widespread relational exposure that is not limited to specific ages or kin types, but instead structured by demographic, health, and kinship regimes across Europe.

Exposure to kin with disability depends on two linked processes: the age composition of kin networks (how many and which kin are alive at each age in a context) and the age gradient of disability (how likely individuals are to experience disability at each age in a context). Mapping exposure to kin with disability helps capture how many individuals are likely to face caregiving responsibilities, which is crucial since such exposure can translate into an often overlooked source of inequalities in individuals' (social and health) outcomes, intersecting and accumulating with the other axes of stratification, such as gender and socioeconomic status. In fact, the exposure to kin disability can also be understood as a hidden structural inequality, placing unequal care-related risks and constraints on individuals that are rarely captured. Demographic and health trajectories shape the likelihood that individuals will face disability caregiving demands across their lives, so

that exposure to kin disability is not merely an individual circumstance but a demographic driver of caregiving pressure.

Our results also speak to future demographic change. Because kinship networks and their health characteristics are structured by fertility, mortality and health regimes, demographic changes systematically redistribute potential care demands across generations and ages. Ongoing trends suggest that exposure to kin with disability is likely to intensify in important ways in the future across Europe. Sustained increases in life expectancy and population longevity act to extend the intergenerational overlaps, so that individuals are increasingly likely to coexist for longer periods with ageing parents and grandparents at ages when disability risk is highest. At the same time, sustained fertility postponement shifts this overlap to even older ages, while persistently low fertility reduces the number of siblings and descendants within families, shrinking the pool of potential caregivers with whom responsibilities can be shared. Future kinship networks are therefore likely to be even more vertically extended but numerically thinner, concentrating disability caregiving demands on fewer individuals and potentially intensifying pressures over the life course. These demographic pressures may be partly counterbalanced if countries succeed in reducing disability risks and postponing disability onset to later ages, in line with a compression of disability (52). Future exposure will therefore depend not only on changes in kinship structure, but also on the extent to which improvements in population health are able to delay or reduce disability at older ages. Italy provides an encouraging example (53), showing how disability-free grandparenthood (the number of years in which individuals are both grandparents and disability-free) has expanded over time. In fact, despite a later transition to grandparenthood, gains in healthy longevity may partly offset the pressures associated with older kin networks by extending the period during which older kin remain alive and disability-free. A similar pattern has been documented in the United States using self-rated health rather than disability, with evidence of increases in the years spent as a grandparent in good health (54).

This work also has some limitations, alongside methodological and interpretive considerations that should be kept in mind. First, we restrict attention to grandparents, parents, siblings, children, and grandchildren, thereby excluding extended kin and partners, even though they may also be important sources of support. Their exposure is therefore relevant to consider, especially for partners at older ages, when disability risk is highest. We do not include the latter here because partners' relationships can not (yet) be considered using standard demographic kinship models in the same way as biological kin. Incorporating partnership histories, union formation and dissolution, and partners' characteristics (such as age, determining their survival) would require a distinct and more complex modelling framework and more in-depth data, likely not

available for all Europe. For this reason, the omission of partners should not be interpreted as indicating lesser substantive importance, but rather as a limitation of the current modelling design. More generally, because affinal kin are excluded, the estimates presented here should be understood as a lower bound on exposure to disability through kinship. Including partners and in-laws would likely increase both the probability of exposure and the absolute number of individuals exposed, for example through older parents-in-law in midlife and older partners in later life. Extending the analysis to partners and wider affinal kin is therefore an important priority for future work. Second, kin are only part of individuals' broader social networks, and our estimates only reflect *potential exposure* to kin disability rather than the actual quality or intensity of these relationships. Moreover, the consequences of having a kin member with disability may depend not only on the presence of disability itself, but also on how many other kin are available within the network. The effect of exposure may differ in smaller versus larger kinship networks, as caregiving responsibilities can be shared or concentrated depending on the number and type of available relatives. While our proportion measure partly captures this relational dimension by expressing kin with disability relative to overall kin availability, the heterogeneous implications of exposure depending on the kin composition, kin characteristics, and the presence of a broader social network remain beyond the scope of our analysis. Third, our estimates implicitly assume that the disability prevalence observed among individuals of a given age also applies to those who are kin of the focal individual and have that age, so that the age profile of disability is applied to the age profile of the focal individual's kin. This assumption may not always hold. For instance, individuals with disability at birth or originating in early life are less likely to have children, and thus disability prevalence among parents and grandparents may be lower than in the general population of the same age. However, disabilities that are present from birth or early life constitute only a (relatively small) share of all disability cases, whereas the vast majority of disabilities arise at older ages as a consequence of age-related health deterioration. Since the onset of most disabilities occurs in later life, and thus after the reproductive period, the potential downward bias in disability prevalence among older kin is likely to be limited. Fourth, the measurement of disability is based on the GALI indicator, which is a self-reported measure of activity limitations. Although GALI is widely used and harmonised across European countries (55–58) and in this study is taken from EU-SILC, where it is administered within a comparable survey framework, cross-national differences in disability prevalence may still partly reflect variation in subjective perceptions of limitations. These perceptions may be influenced by contextual factors such as infrastructure, accessibility, social norms, and expectations regarding functioning. As a result, individuals with similar objective health conditions may report different levels of limitation depending on the environments in which they live. At the same time, within the

ICF framework (2), disability is exactly defined as the interaction between individual characteristics and contextual factors. Such variation in perception is thus not merely a source of measurement bias but may capture the ICF-defined disability, which reflects how inclusive or constraining different contexts are. Therefore, our estimates may incorporate both differences in underlying objective health conditions and differences in how environments shape the experience of activity limitations, which can be considered an integral part of the exposure we aim to capture. Finally, the results are presented for women only. We focus on women because exposure to kin disability is especially relevant for gender inequality, given women's disproportionate involvement in unpaid family caregiving. The same modelling approach can be applied to men, and we expect the age profiles of the probability of having at least one kin with disability and the proportion of kin with disability to be broadly similar for men and women. This is particularly the case for ascendant kin, since parents and grandparents are structurally the same for male and female focal individuals. Population-level absolute numbers would increase substantially if both women and men were included, and as male and female population sizes are broadly similar at most ages, including men would approximately double the total number of individuals exposed (although not exactly, given gender differences in population age structures and survival). Similarly, the disability-kinship multiplier would also be around doubled when exposure is measured for both women and men.

These results carry substantial policy implications. The broader social implications associated with having kin with disability arise not from disability per se, but from the disadvantages produced by non-inclusive institutions, inaccessible environments, discriminatory norms, and care systems. Through kinship, these disadvantages extend far beyond those directly experiencing disability, creating a multiplicative effect across family networks. Exposure to kin disability is thus a demographic pathway through which inequality can accumulate. Policies should therefore adopt a more relational perspective on disability, recognising its indirect effects across generations and kinship networks. Strengthening integrated support systems that address not only persons with disability but also their kin members is essential to reducing cumulative social and economic disadvantages associated with long-term care responsibilities. At the same time, as the future exposure to disability within the kinship network will also depend on whether disability can be delayed or reduced at older ages, investments in health systems, prevention, healthier behaviours, medical research, technological innovation, and more accessible environments may compress disability into later life, thereby reducing exposure to kin with disability over part of the life course. Policies that support person-centred life projects for young people with disability, such as independent living and labour-market participation, can improve their lives while also reducing constraints on families and limiting the concentration of care responsibilities within households.

Likewise, healthy ageing strategies, assistive technologies, and accessible communities can help older adults maintain autonomy for longer, with positive effects both for them and their kin. Finally, the negative effects of exposure to kin with disability are likely to fall disproportionately on women, given persistent gender norms in care provision and existing inequalities in the labour market in many countries. Exposure to kin disability can contribute to reduced labour force participation, interrupted career trajectories, lower lifetime earnings, and pension gaps, thereby reinforcing cumulative gender inequality over the life course. Addressing these dynamics requires policies aimed at expanding accessible and affordable long-term care services, strengthening caregiver allowances, and ensuring pension credits for caregiving periods, to help mitigate the long-term socioeconomic consequences of extended family-based care. The meaning and consequences of exposure are unlikely to be uniform across countries. The same level of exposure to kin disability may translate into different caregiving pressures depending on how care is organised between families, public institutions, and market providers (34–38). In more familistic contexts, where care responsibilities remain concentrated within households, having a close kin member with disability may generate stronger demands on family members. In more de-familialised contexts, where long-term care services, formal childcare, income support, and other public or market-based provisions are more developed, these pressures may be partly mitigated. Differences in exposure to kin disability should therefore be interpreted not only as the result of demographic histories and health profiles, but also in relation to care regimes that shape whether disability within kinship networks becomes a source of concrete caregiving responsibilities and social inequality. The policy measures highlighted above are, thus, particularly relevant in more familistic regimes, where care is more concentrated within households and where women's outcomes, opportunities and decisions remain closely intertwined with unpaid caregiving responsibilities, especially when non-inclusive institutions, inaccessible spaces, and discriminatory norms amplify the consequences of disability across family networks and generations. Finally, beyond these substantive implications, the estimates generated in this study provide a new empirical basis for policy and future research. By quantifying the distribution of exposure to kin disability by age, gender, kin type (and country), the study makes visible the size and composition of the populations exposed to disability through kin networks, information that has so far been largely unavailable. These estimates are provided as .csv files in the public reproducibility repository (see Methods section), together with the code used to compute and instructions on how to export them. These files can serve to inform the design and targeting of care systems, caregiver support, and healthy ageing policies, while also providing a baseline for future work on the social and economic consequences of kin disability.

More broadly, the study contributes empirically, conceptually, and methodologically to existing research. Empirically, it provides comparable demographic estimates of exposure to kin disability across Europe, quantifying both how widespread this exposure is and its intensity. In doing so, the measures proposed in this study complement existing indicators of kin availability by incorporating the health dimension of kin, thereby offering a more comprehensive assessment of intergenerational dependency structures. Conceptually, it advances a shift in how disability is understood, not only as an individual experience, but also as a relational exposure that extends through kinship networks. These measures may also be interpreted as indicators of kinship vulnerability, capturing how "care-demanding" kinship networks are likely to be, for whom, and for how many people. Methodologically, the study shows that model-based estimates of kin disability exposure remain stable across alternative specifications and approximate observable empirical evidence where direct comparison is possible.

Future studies are called to further examine the drivers of changes in exposure to kin with disability over time and across countries, including the different roles of fertility, mortality, and disability prevalence.

Materials and methods

Data

The analytical approach is based on kinship matrix models, where the input data to construct kinship networks are fertility and mortality rates and, in this case, disability prevalence.

Age and gender specific mortality rates, female age specific fertility rates, and the sex ratio at birth come from the United Nations World Population Prospects (UNWPP) (59) with data from 1950 to 2023. While female age-specific fertility rates are routinely available in international demographic databases, male fertility schedules are considerably less consistently reported. Rather than relying on the androgynous approximation (i.e., assuming identical fertility schedules for men and women (41)) as in similar works, we constructed male fertility rates drawing on different sources. For country estimates as reported in the appendix, whenever available, we used observed male age-specific fertility rates from the Human Fertility Collection (60); for countries and years not covered by the HFC, we derived male fertility schedules by shifting the female age pattern according to the country-specific difference between the mean age at fatherhood and the mean age at childbearing, following Schoumaker (61). Appendix, Table S3 summarises, for each country, the periods covered by each fertility data source. For the European average (EU_27) we use the

androgynous approximation. Fertility and mortality rates by gender and age from 1950 to 2023 in selected countries are shown in the Appendix, Figures S13.1 to S13.3.

In order to compute the (population-level) absolute number of individuals exposed to disability, we draw on the total population count of Europe, by gender and age, of 2021 from UNWPP.

Age and gender specific disability prevalence are derived from the EU Statistics on Income and Living Conditions (EU-SILC), including the 2021 ad hoc module, which collects information on activity limitations through the Global Activity Limitation Indicator (GALI) for individuals of all ages (with children covered in the ad hoc module). The GALI question asks respondents whether, for at least the past six months, they have been limited in activities people usually do because of health problems. Response categories include "severely limited," "limited but not severely," and "not limited", and we classify individuals reporting either severe or moderate limitations as having a disability, while those reporting no limitations as disability-free. Please see Appendix (Survey research disclosure: EU-SILC/GALI) for further information. The GALI constitutes the official measure used by Eurostat to compute the Healthy Life Years (HLY) indicator (one of the core indicators of the European Structural Indicators and the European Pillar of Social Rights), and it has been validated for its goodness in terms of harmonization and comparability of health measurement across age groups, periods, and European countries (55–58). Disability prevalence from the SILC survey is available for most European countries and a European average, and it is also publicly available from Eurostat websites, detailed by age classes (https://doi.org/10.2908/HLTH_SILC_12 and https://doi.org/10.2908/ILC_HCH13). This public data is available by gender for ages above 16. For younger ages, gender-specific data are not available, and therefore, males and females aged 0–16 are assigned to have the same disability prevalence (i.e., equal prevalence across genders). Age and gender specific prevalence used are shown in Appendix, Figure S14.1 and S14.2.

We draw on aggregate EU-27 estimates based on the standard 27-country European Union definition. This aggregate is available consistently across all data sources used in the analysis, including fertility, mortality, and population counts from the UNWPP and disability prevalence from Eurostat website. This EU_27 aggregate is used to present overall European estimates, alongside the country-specific results reported in the appendix. Input data at EU-27 level are calculated by Eurostat and UNWPP as the population-weighted arithmetic average of individual national figures (62).

Finally, for robustness analysis comparing mothers, fathers, and children who live in the same household as the focal individual with equivalent model's results, we draw on individual-level data from EU-SILC survey (DOI: 10.2907/10.2907/EUSILC20042024REV1).

Methods

Our analysis relies on formal demographic kinship models that reconstruct the expected number and composition of living kin for individuals at each age using observed fertility and mortality schedules (from 1950 to 2021). This approach follows the demographic principle that a population's age-specific vital rates implicitly determine its kinship structure (Goodman et al., 1974), as kinship networks are endogenous products of demographic regimes. By combining age and gender specific fertility and mortality rates over time, it is therefore possible to derive the expected number of each type of kin at each age of a focal individual (ego).

We employ the two-sex, time-varying matrix kinship framework developed by Caswell and colleagues (41, 42). This method allows for modelling both male and female populations of kin (*two-sex*), and accounts for historical changes in timing and trends of fertility and mortality (*time-varying*), so that kin availability reflects past demographic regimes. All estimates are computed separately for female and male focal individuals, and are here shown only for women. We implement the kinship matrix computations using the DemoKin package in R (64).

To incorporate disability into the kinship framework, we extend the kinship model (similarly to Feng et al. (2024)), by applying the age and gender specific prevalence of disability. The kinship matrix method, for each age of the focal, produces the expected number of living kin of each type, age, and gender; applying the age and gender specific disability prevalence to these modelled age distributions of kin allows to specify the expected number of kin by disability state, yielding the expected number of kin with disability and the expected number of kin without disability. This assumes that disability prevalence among kin by age and gender corresponds to population-level prevalence for that group.

From these quantities, we compute three complementary measures of exposure to disability within the kin network:

1. *Probability of having at least one kin with disability*: For each kin type, this measure represents the probability that an individual of a specific age has at least one living kin of that type with disability. For example, the probability of having at least one grandparent with disability corresponds to the likelihood that a focal individual, at a specific age, has one or more grandparents with disability. This measure captures how widespread exposure is across individuals in the population, that is, the share of individuals at a given age who are likely to experience the exposure to disability within a specific kin relationship during their life course.

2. *The absolute (population-level) number of women having at least one kin with disability*: For each kin type and at each specific age, we compute the absolute number of women, at the population level in Europe, who have at least one kin with disability.
3. *Proportion of kin with disability*: This measure is computed as the expected number of kin with disability divided by the total expected number of (alive) kin at each age of the focal individual. It is calculated for the overall kinship network and also separately by kin type. This indicator reflects the intensity of exposure to disability within the kinship network and can be interpreted as a measure of the weight of potential caregiving demand faced by individuals in their kinship network.

The probability of having at least one kin with disability, by kin type, is derived from the expected number of kin with disability. As in Song & Mare (65), we assume that the number of kin follows a Poisson distribution with mean $\lambda = \mu(x)$, so that $\mu(x)$ denotes the expected number of kin at the age x of the focal. If N is the random variable representing the number of kin, then:

$$N \sim Poisson(\lambda = \mu(x))$$

so that

$$P(N = n) = \frac{\mu(x)^n}{n!} e^{-\mu(x)}, n = 0,1,2,\ldots$$

The probability of having no kin is defined as:

$$P(N = 0) = e^{-\mu(x)}$$

and the probability of having at least one kin is defined as:

$$P(N \geq 1) = 1 - e^{-\mu(x)}$$

Feng et al. (46) extended this logic to specify kin's health condition. Let $\pi(x)$ denote the age-specific prevalence of disability, so that the expected number of kin with disability, for each kin type, is defined as:

$$y(x) = \pi(x)\, \mu(x)$$

Under the Poisson assumption, and if M is the random variable representing the number of kin with disability, the probability of having at least one kin with disability is defined as:

$$P(M \geq 1) = 1 - e^{-y(x)}$$

This approach assumes that events have a fixed mean $y(x)$ equal to the variance. In real kinship, however, disability risk may cluster within families because of shared genetic, socioeconomic, or environmental factors, and some focal individuals may belong to systematically higher-risk kin networks than others. Such mechanisms may induce overdispersion, thereby violating the assumption of the variance being equal to the mean, as the variance of the count of kin with disability is inflated beyond the Poisson one.

To account for potential overdispersion, we extend the previous formulation and assume instead that the number of kin with disability follows a Negative Binomial distribution (Poisson-Gamma mixture), relaxing the constraint of the Poisson model.

Conditional on a latent rate Λ,

$$M \mid \Lambda \sim Poisson(\Lambda)$$

and the rate varies across focal individuals:

$$\Lambda \sim Gamma(r, \theta)$$

The number of kin with disability M follows a Negative Binomial with mean $y(x)$ and variance:

$$Var(M) = y(x) + ky(x)^2, \text{ where } k = \frac{1}{r}$$

The probability of having no kin with disability, for each kin type, is thus defined as:

$$P(M = 0) = (1 + ky(x))^{-\frac{1}{k}}$$

and therefore the probability of having at least one kin with disability, for each kin type, is:

$$P(M \geq 1) = 1 - (1 + ky(x))^{-\frac{1}{k}}$$

As $k \to 0$, i.e. in the case of no overdispersion, this expression converges back to the Poisson definition of variance equal to the mean, and $P(M \geq 1) = 1 - e^{-y(x)}$, while a larger k value indicates stronger overdispersion. In an overdispersed process, the variance of counts is higher; that is, some focal individuals have many kin with disability, but more focal individuals have no kin with disability, and the fraction of those having at least one kin with disability is therefore smaller than from the Poisson assumption. We adopt a conservative baseline level of $k = 0.3$, which represents moderate overdispersion and captures possible clustering within families. As robustness, we report sensitivity checks with $k \in \{0 \text{ to } 0.6\}$, in the Appendix Figures 7 and 8. Other methodological considerations in the computation of the probability are discussed in the Appendix (section “Methodological note for the probability computation”).

To compute the absolute number of individuals having at least one kin with disability, we multiplied the computed probability estimates by the total population count in 2021 at each (focal) age for the Europe aggregate.

These computations require as input: age and gender-specific mortality and fertility rates over time; sex ratios at birth; gender of the focal individual (male or female kinship networks); the age and gender-specific total population (of 2021) for Europe; the gender-specific prevalence of disability for a given time point of interest (2021). Since the information on the prevalence is not publicly available for all single ages (rather, they are available by age classes), generalised additive models (66) and penalised regression splines (67) are used to model the trend of the prevalence by single age (with the number of knots selected to balance flexibility and avoid overfitting, and with

smoothness estimated using restricted maximum likelihood). Smoothed and unsmoothed prevalence over ages are presented in the Appendix, Figures S14.1 and S14.2.

All analyses are performed in the R software (68) and documented in an open-access and reproducible repository (which will be available in the published version of this paper).

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Supporting Information (Appendix)

Probability of having at least one kin with disability by country

Countries slightly differ in the overall level, shape, and timing of these probability profiles by age. In some contexts, probabilities are generally higher across kin types (e.g. Portugal), whereas in others they remain lower (e.g. Italy). In general, cross-country differences generally shrink at older ages in the case of ascendant kin and siblings (as they are typically extinct from the kin networks), while they narrow for siblings and descendants (likely because of heterogeneity in fertility levels and thus the number of descendant kin overall). The curves also vary in how strongly exposure is concentrated at particular ages, as some are more clearly peaked, while others are flatter and more gradual, with smoother shifts across kin types and ages.

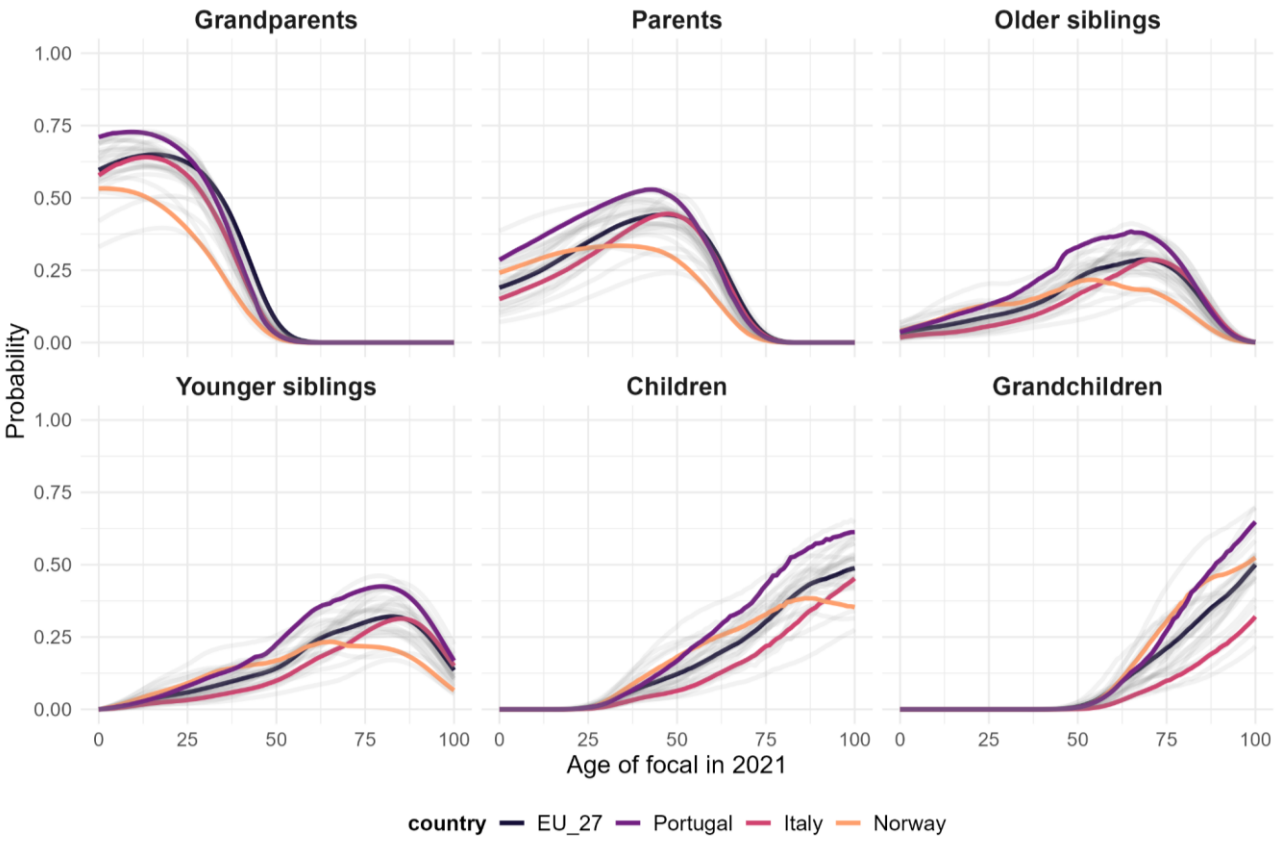


Figure S1.1: Probability of having at least a kin with disability, by kin type, age of a focal woman and European countries in 2021; The probability for Europe, Portugal, Italy, and Norway are highlighted

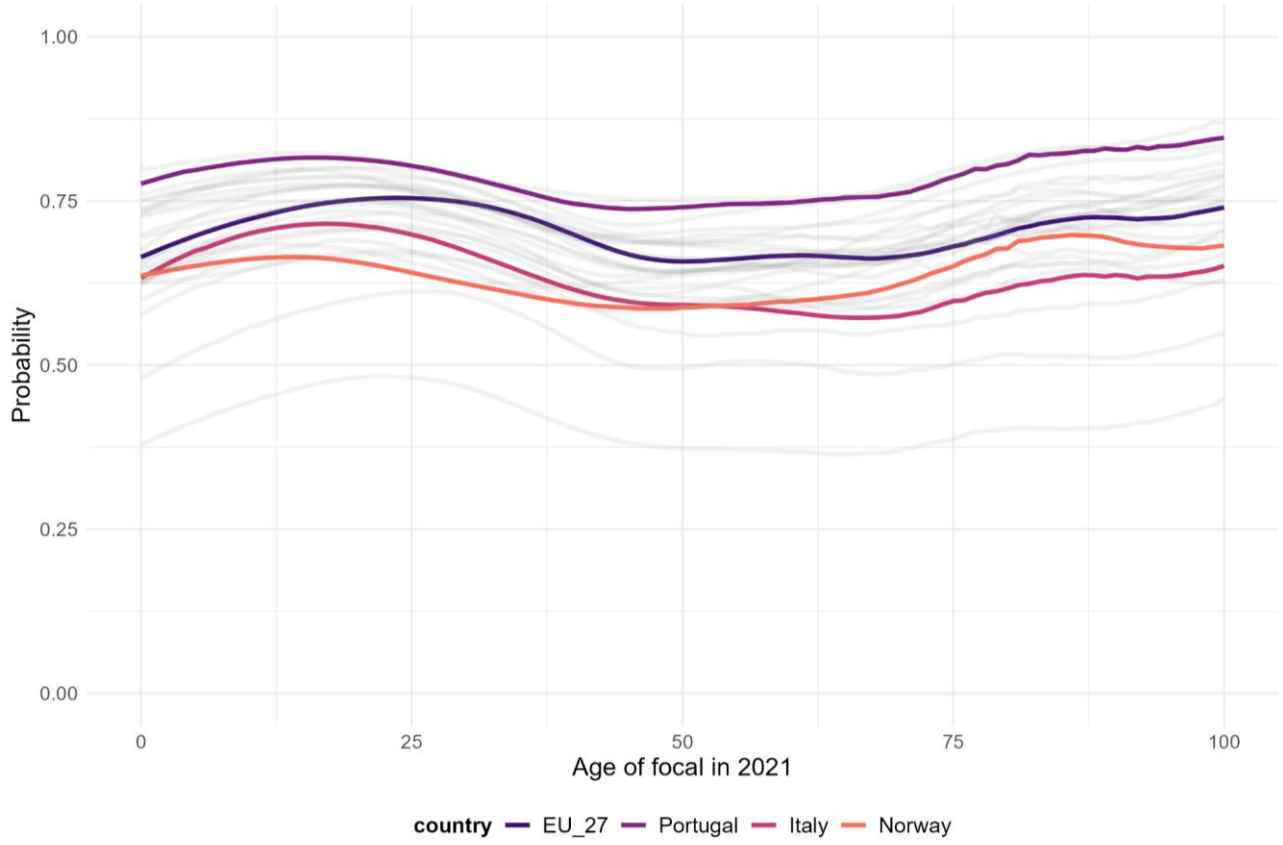


Figure S1.2: Probability of having at least a kin with disability, by age of a focal woman and country in 2021. The probability for Europe, Portugal, Italy, and Norway are highlighted

Absolute number of women having at least one kin with disability by country

As a country example, in Italy, exposure to grandparent disability involves up to around 175,000 women during childhood and early adulthood. Exposure to parental disability becomes most prominent in midlife, reaching more than 200,000 women between ages 40 and 55, and representing the largest group of exposed women across ages. Exposure to older siblings with disability generally involves up to around 100,000 women, peaking in later adulthood and early older age. At older ages, exposure to younger siblings can affect up to around 80,000 women, while exposure through children and grandchildren can involve up to about 75,000 and 35,000 women, respectively.

The overall age profile and sequencing of kin types are similar across different contexts. Although this common sequencing holds across all countries, the curves still slightly differ in overall levels, in how strongly exposure is concentrated at particular ages, and in the precise timing at which different kin types become most important. In some countries (e.g. Italy), the profiles are

generally higher and more sharply peaked, with exposure more strongly concentrated in early life and midlife. In others (e.g. Sweden), the curves are lower overall and flatter, indicating a more gradual age pattern and less pronounced concentration at specific ages. Countries also differ in how clearly one kin type gives way to another as the main source of exposure. In some contexts (e.g. Poland), the life-course shift from grandparent-driven exposure in early life to parent-driven exposure in midlife, and later to sibling- and descendant-driven exposure, is more clearly marked. In others (e.g. Germany), these phases overlap more, so that the decline of one kin type and the rise of another occur more gradually and less suddenly. Differences across countries in the absolute number of women exposed reflect not only differences in the probability of exposure, but also the size and age profile of the underlying female population. In other words, the number exposed at each age depends both on the likelihood of having at least one kin with disability and on how many women are alive at that age group in a given country. Variation in the magnitude, shape, and location of these curves thus reflects also differences in cohort size across ages, shaped by each country's demographic history, including the timing and intensity of baby booms, subsequent fertility decline, and broader processes of population ageing.

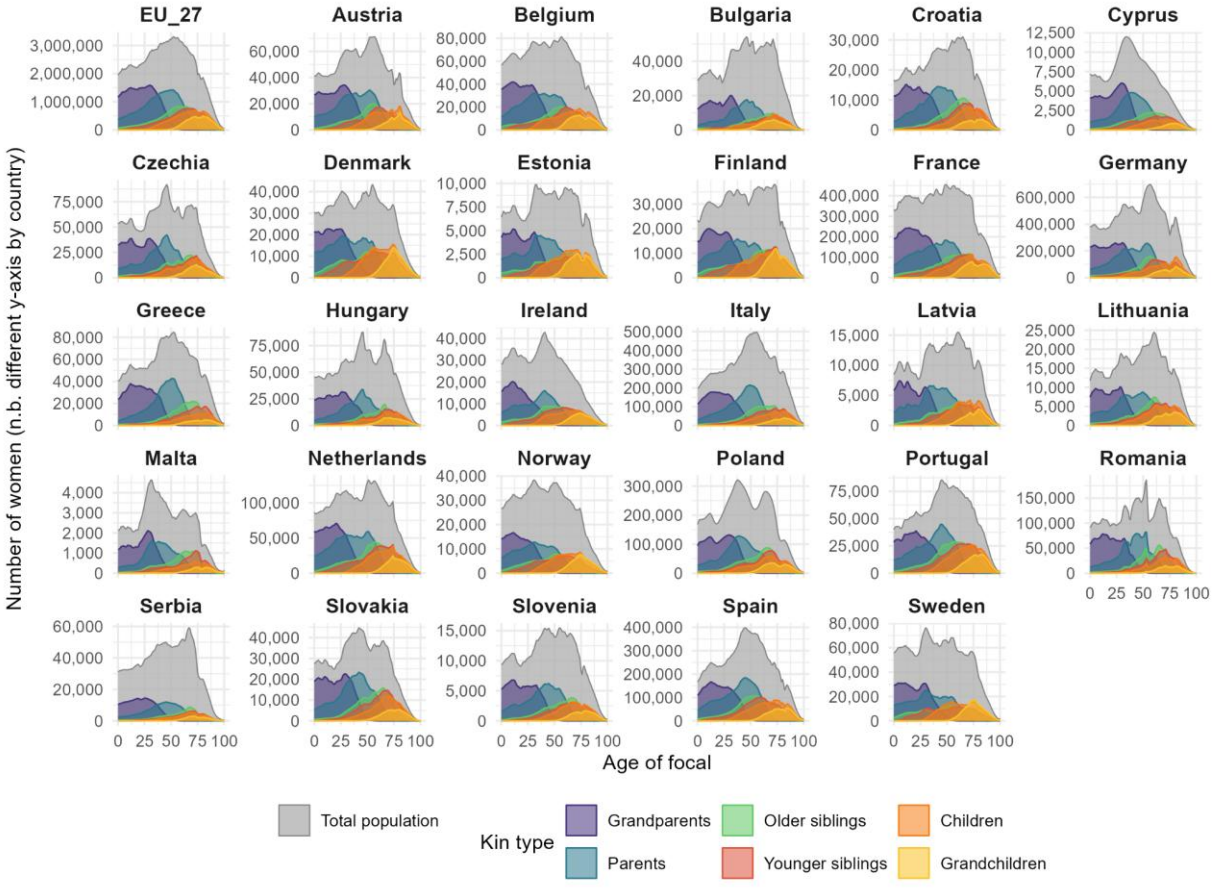


Figure S2: Absolute number of women having at least one kin with disability, by type of kin, age of the focal woman and country, compared to the total population of women at each age (grey areas)

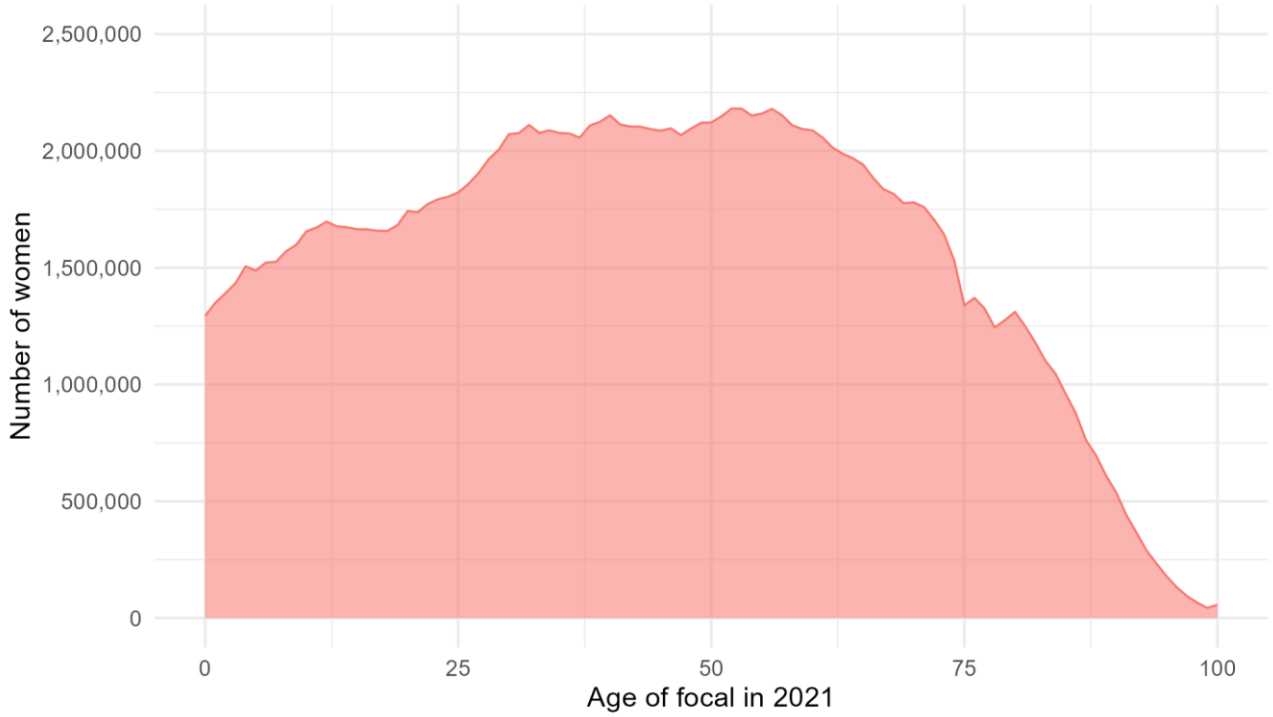


Figure S3: Absolute number of women having at least one kin with disability, by age of the focal woman in European Union (EU: 27)

Table S1: Total absolute number of women having at least one kin with disability, by country, bounded using the same range of k values used in the computation of the probabilities

Country	Exposed population, millions (lower-upper bounds)	% of overall national population (corresponding to the probability measure)
European Union (EU: 27)	**159.76 (147.86-174.62)**	**69.8**
Austria	3.25 (3.01-3.56)	71.6
Belgium	4.04 (3.74-4.41)	69.0

Bulgaria	1.89 (1.77-2.04)	53.0
Croatia	1.47 (1.36-1.61)	72.2
Cyprus	0.44 (0.41-0.48)	68.1
Czechia	3.68 (3.41-4.03)	68.9
Denmark	2.31 (2.13-2.52)	78.5
Estonia	0.51 (0.47-0.56)	73.0
Finland	2.05 (1.90-2.25)	73.3
France	23.44 (21.70-25.61)	68.9
Germany	27.19 (25.24-29.63)	64.2
Greece	3.59 (3.33-3.92)	65.3
Hungary	3.01 (2.81-3.27)	59.5
Ireland	1.67 (1.55-1.83)	66.3
Italy	19.18 (17.82-20.87)	62.7
Latvia	0.71 (0.66-0.78)	69.9
Lithuania	1.02 (0.95-1.12)	69.0
Malta	0.16 (0.15-0.17)	64.0
Netherlands	6.62 (6.12-7.24)	74.4
Norway	1.67 (1.55-1.82)	62.6
Poland	13.01 (12.06-14.19)	66.3
Portugal	4.21 (3.90-4.61)	77.4
Romania	7.05 (6.52-7.71)	70.8
Serbia	1.47 (1.39-1.56)	40.8
Slovakia	2.13 (1.97-2.33)	76.2
Slovenia	0.67 (0.62-0.73)	63.6
Spain	17.58 (16.26-19.23)	72.4
Sweden	3.25 (3.01-3.53)	62.9

Note: Counts are shown in millions and rounded to two decimals. Shares are percentages of the overall population and rounded to one decimal. Parentheses report lower and upper bounds. Percentages are calculated as the affected population divided by total population (and rounded to one decimal).

Table S2: Disability-kinship multiplier: for each person with disability, how many women (on average) are exposed to kin disability

Country	Women with at least 1 kin with disability, millions	People with disability, millions	Exposed women per person with disability
European Union (EU: 27)	**159.76**	**98.18**	**1.6**
Austria	3.25	2.17	1.5
Belgium	4.04	2.46	1.6
Bulgaria	1.89	1.07	1.8
Croatia	1.47	1.11	1.3
Cyprus	0.44	0.20	2.2
Czechia	3.68	2.25	1.6
Denmark	2.31	1.77	1.3
Estonia	0.51	0.38	1.3
Finland	2.05	1.39	1.5
France	23.44	12.62	1.9
Germany	27.19	17.31	1.6
Greece	3.59	2.17	1.7
Hungary	3.01	1.80	1.7
Ireland	1.67	0.80	2.1
Italy	19.18	11.69	1.6
Latvia	0.71	0.58	1.2

Lithuania	1.02	0.74	1.4
Malta	0.16	0.08	2.1
Netherlands	6.62	4.55	1.5
Norway	1.67	0.88	1.9
Poland	13.01	7.27	1.8
Portugal	4.21	3.19	1.3
Romania	7.05	4.82	1.5
Serbia	1.47	0.72	2.0
Slovakia	2.13	1.42	1.5
Slovenia	0.67	0.40	1.7
Spain	17.58	11.95	1.5
Sweden	3.25	1.76	1.8

Note: Counts are shown in millions and rounded to two decimals. The count of people with disability is computed by multiplying the (age-gender-specific) prevalence by the population counts. The multiplier is calculated as the number of women with at least one kin with disability divided by the total number of people with disability, and is rounded to one decimal.

Proportion of kin with disability by country

In terms of country differences, in Portugal, compared to the other two countries, grandparents have the largest share of the total proportion of kin with disability at younger ages, and children and grandchildren at older ages. In Italy, the contribution of parents is relatively higher around midlife, while at older ages, descendants play a smaller role than in Portugal. In Norway, the distribution is flatter overall, with the proportion of kin with disability remaining at less than 20% across all ages.

Portugal has the highest proportion in Europe, while Italy has close to average levels, and lower than the other Southern European countries. France and Ireland show relatively low proportions (likely reflecting the overall larger number of kin in these countries due to higher fertility). Within Northern Europe, Finland and Denmark are exceptions, showing levels that are closer to the Baltic states (Estonia, Latvia, Lithuania) than to their Nordic neighbouring countries. Eastern Europe and the Balkanic area are quite heterogeneous, with both low (Serbia and Bulgaria) and high (Croatia) values depending on the country.

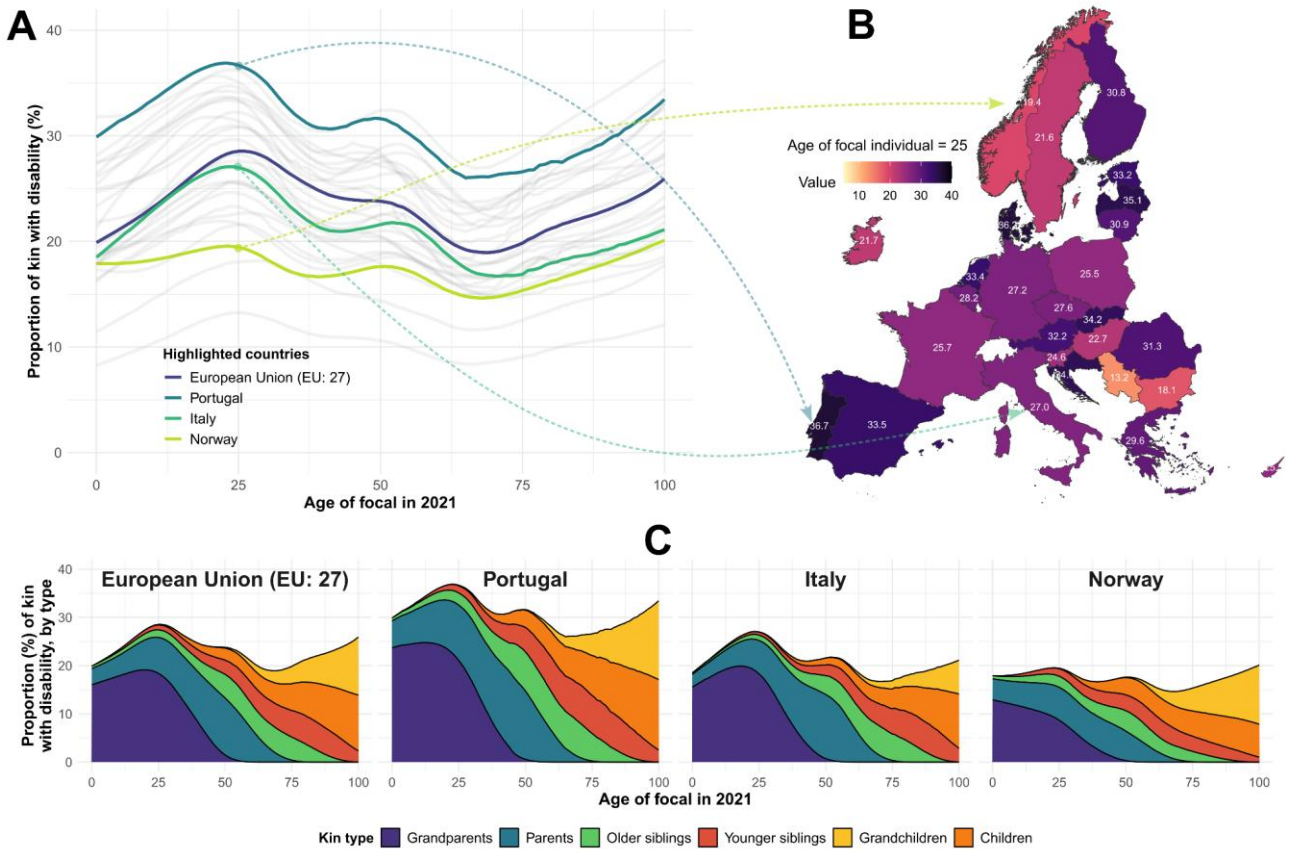
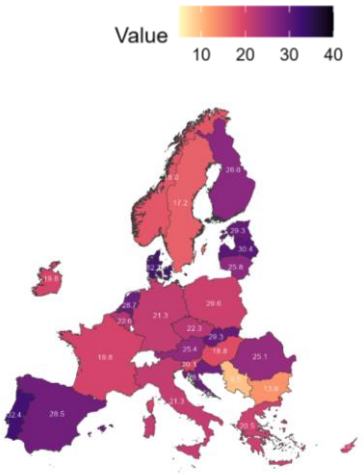


Figure S4: Magnitude of the exposure measured using the proportion of kin with disability. Panel A: Proportion of kin with disability over the life of a woman in 2021 in European countries, with values for Europe (EU_27), Portugal, Italy, Norway highlighted; Panel B: Map of the proportion of kin with disability for a 25-year-old woman in 2021 in European Countries. Panel C: Specification of the proportion by kin type, over the life of a woman in 2021 in selected countries.



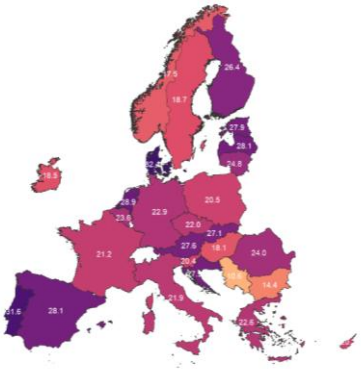
AGE 10

Figure S5.1: Proportion of kin with disability for women in 2021; focal age 10.



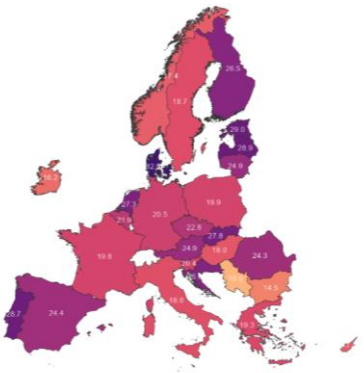
AGE 40

Figure S5.2: Proportion of kin with disability for women in 2021; focal age 40.



AGE 50

Figure S5.3: Proportion of kin with disability for women in 2021; focal age 50.



AGE 85

Figure S5.4: Proportion of kin with disability for women in 2021; focal age 85.

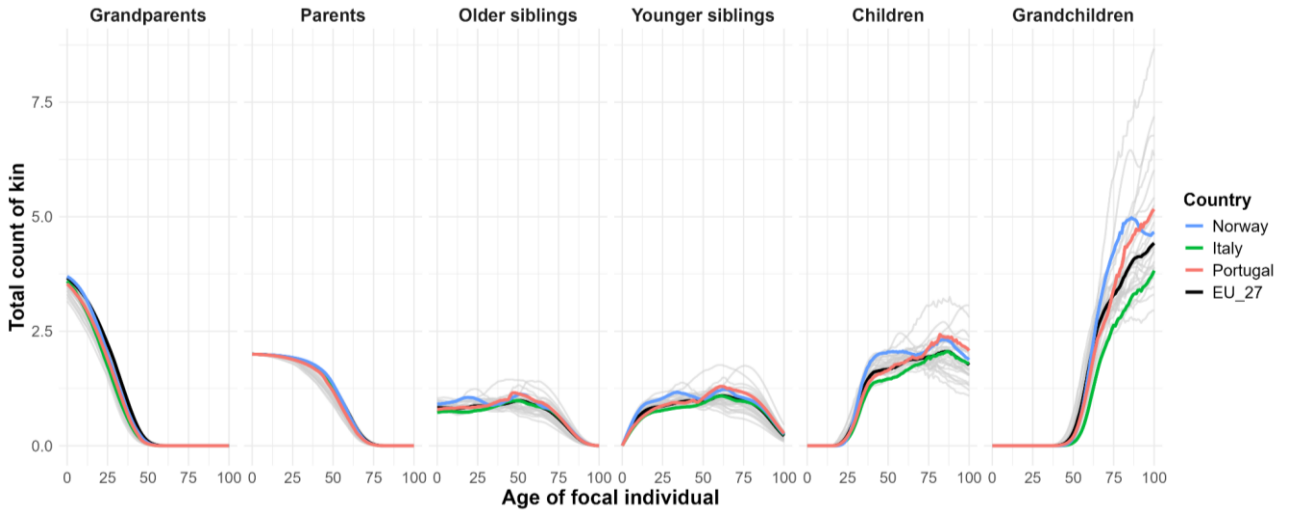


Figure S6.1: Total number of kin by kin type and focal age, for women in 2021, with Europe (EU_27), Portugal, Italy, Norway highlighted

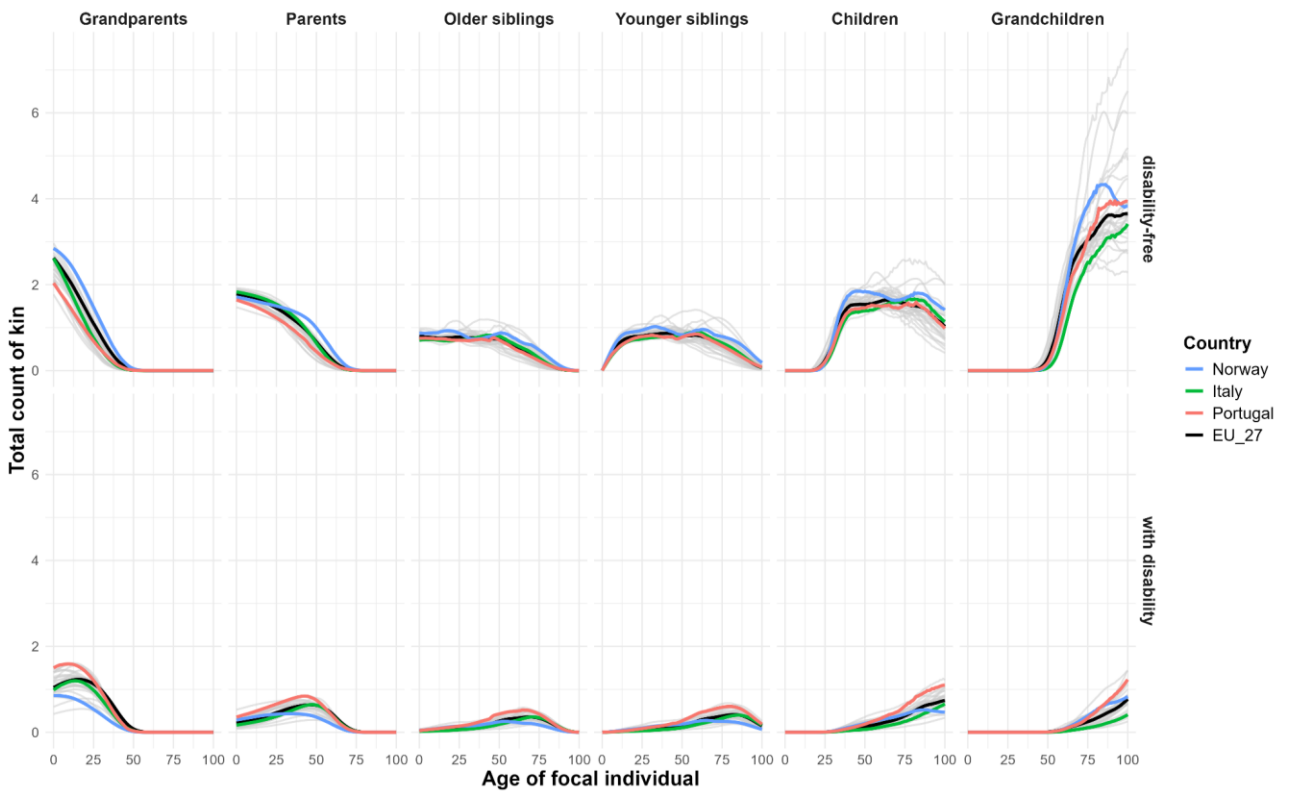
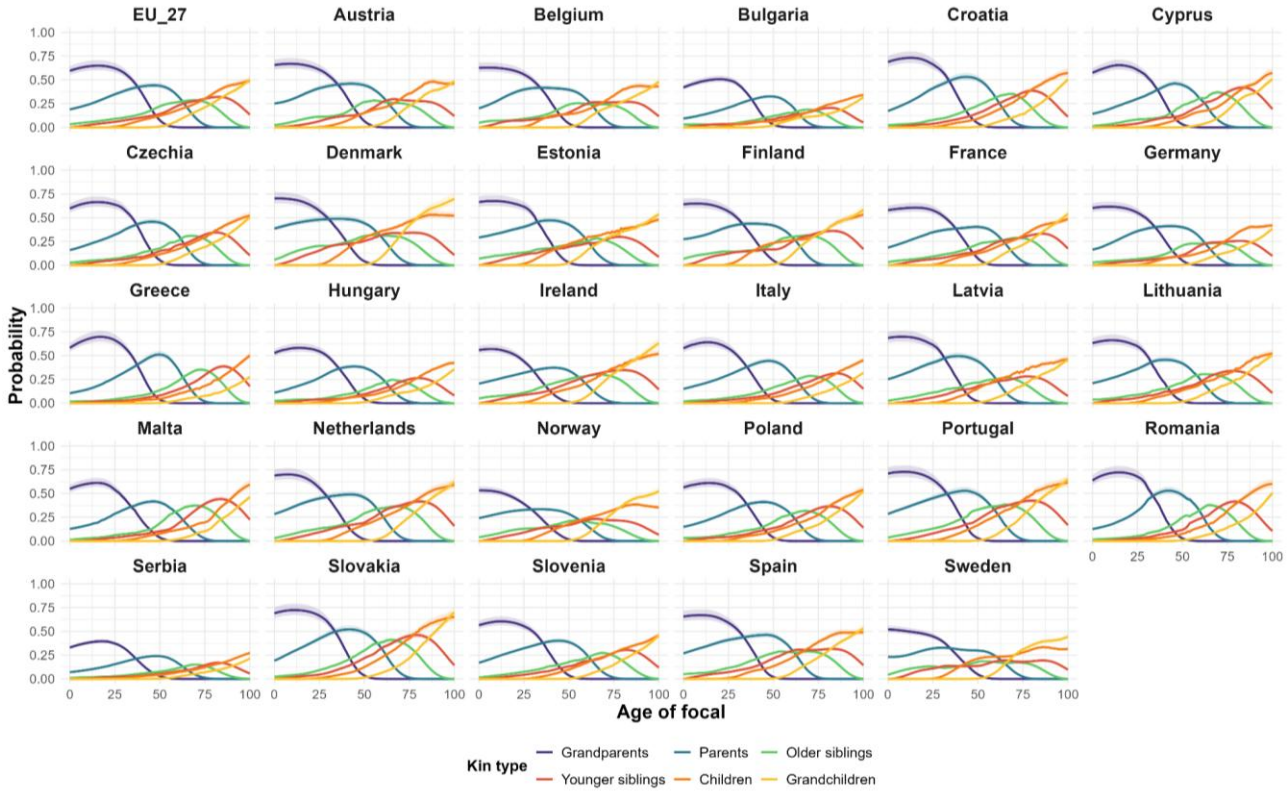


Figure S6.2: Total number of kin by kin type, focal age and kin disability state, for women in 2021, with Europe (EU_27), Portugal, Italy, Norway highlighted

Robustness checks

A



B

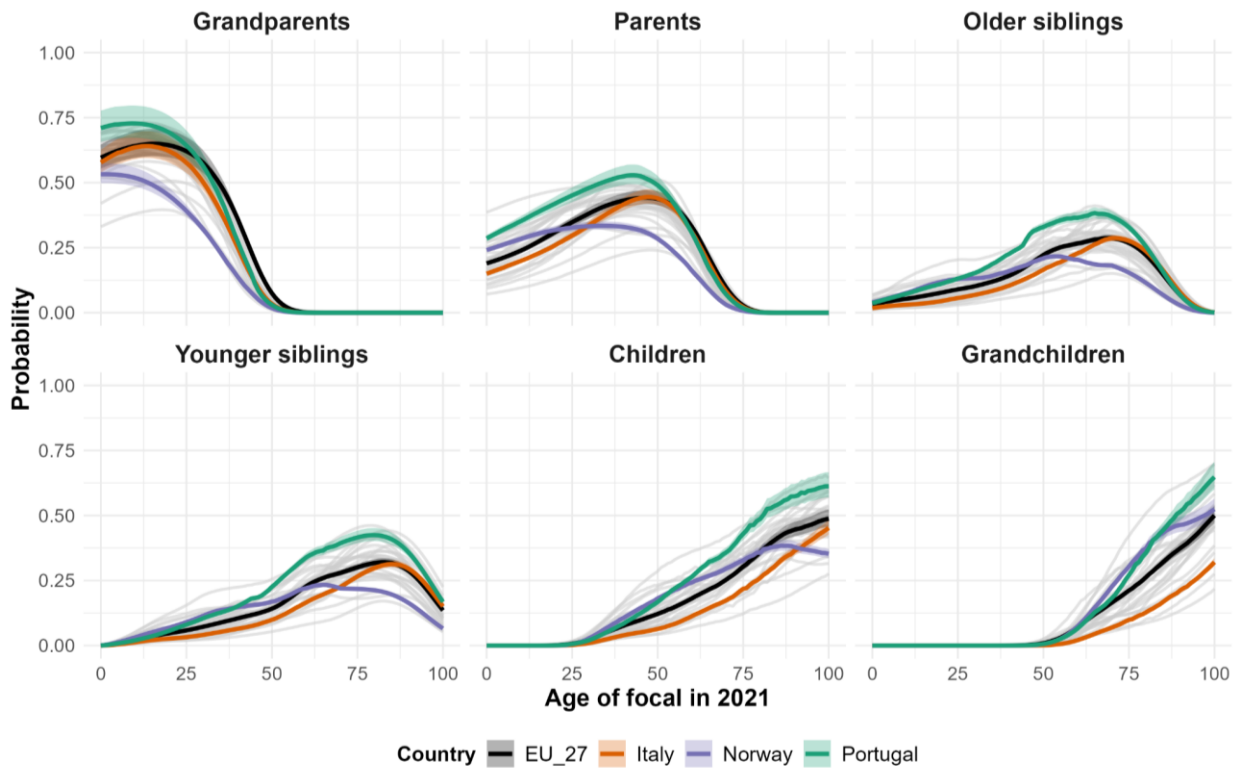


Figure S7: Panel A and B: Probability of having at least a kin with disability, by kin type, age of a focal woman and country in 2021, using a range of k values: from k = 0 (i.e. using the Poisson approximation, meaning no overdispersion) to k = 0.6 (high overdispersion); central estimate corresponding to k=0.3; Europe (EU_27), Portugal, Italy, Norway highlighted

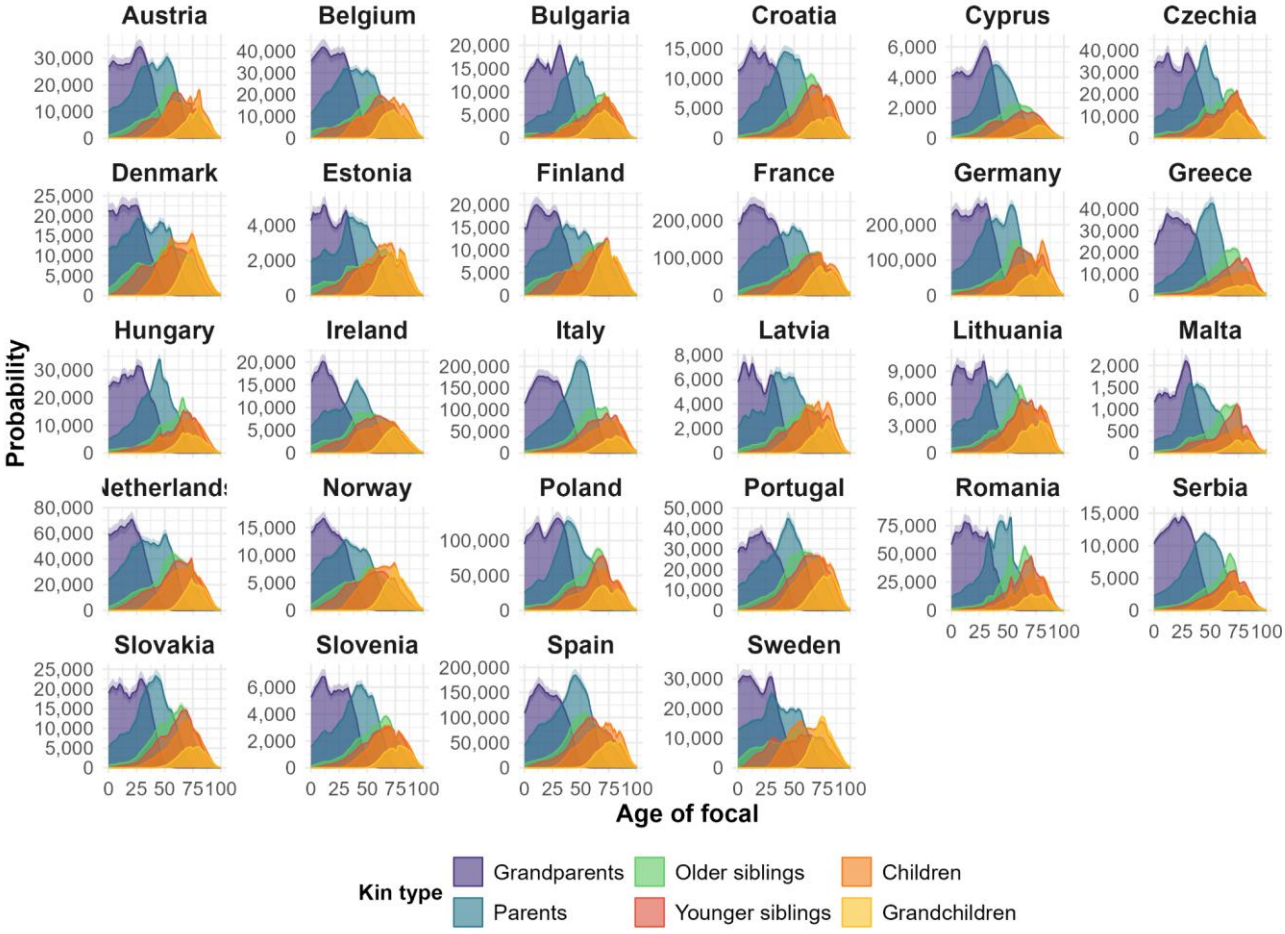


Figure S8: Absolute number of women having at least one kin with disability, by type of kin, age of the focal woman and country, bounded using the same range of k values used in the computation of the probabilities.

Comparison of some of our estimates with SILC data

Here we compare selected estimates derived from our model with corresponding indicators computed from raw SILC data, focusing in particular on the presence of mothers, fathers and children with a disability living in the same household as the focal individual. We selected only this

kin for the comparison because other kin types (such as grandparents) are less likely to live in the same household, and would not thus be captured in SILC data where kin linkage are possible only for individuals living in the same household (e.g. it is not possible to identify the grandparent of an individual if they are not living in the same household), as the kinship structure linkage is not extended outside the household.

Methodological note for the probability computation: For mother and father the total count of this kin for a focal individual is structurally bounded between 0 and 1. There may be cases for individuals having more than one mother or father, e.g. female/male same-sex families, but their prevalence is still quite low, and the SILC data still captures just one mother/father ID, so individuals in this data can only have a maximum of one mother and one father. Although this count is thus binary (0 or 1), we estimate it using the same Negative Binomial distribution used for other kin counts in order to maintain consistency across kin types. Importantly, this specification is unlikely to induce important distortion, because the mean parameter of this distribution (here being the mean of mother/father with disability) is relatively low. When the mean is small, the Negative Binomial distribution concentrates almost all its probability mass between 0 and 1, so that the distribution closely approximates a Bernoulli process.

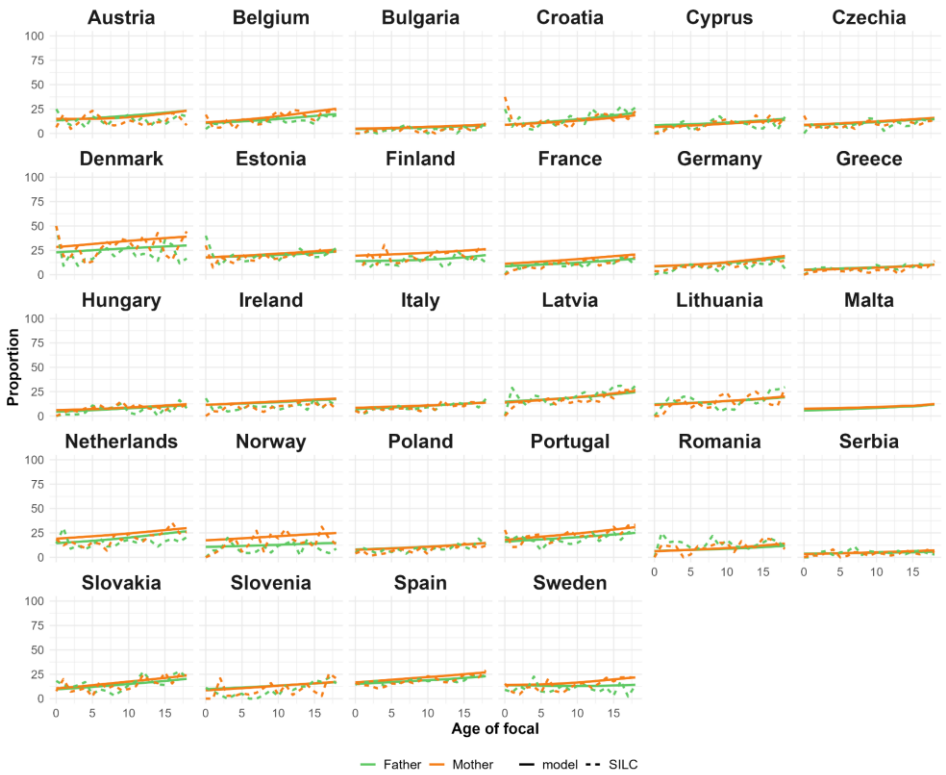


Figure S9: Proportion of parents with disability - comparison between the kinship matrix models and "empirical" ones from SILC data (based on co-resident kin)

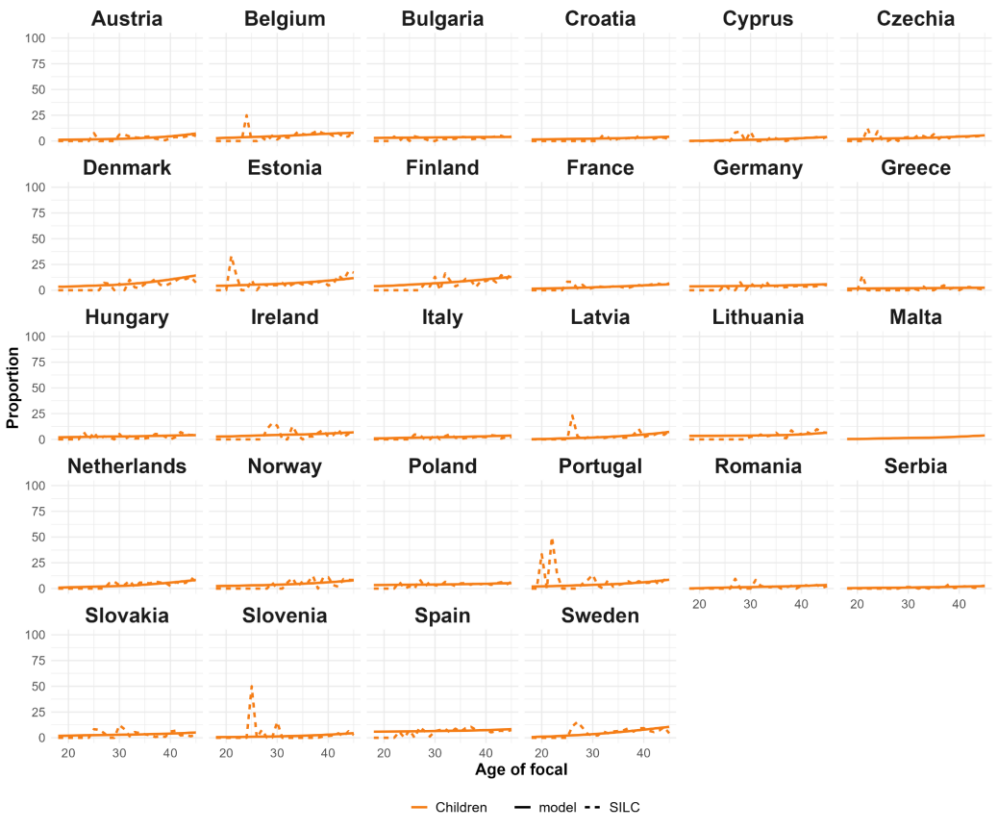


Figure S10: Proportion of children with disability - comparison between the kinship matrix models and "empirical" ones from SILC data (based on co-resident kin)

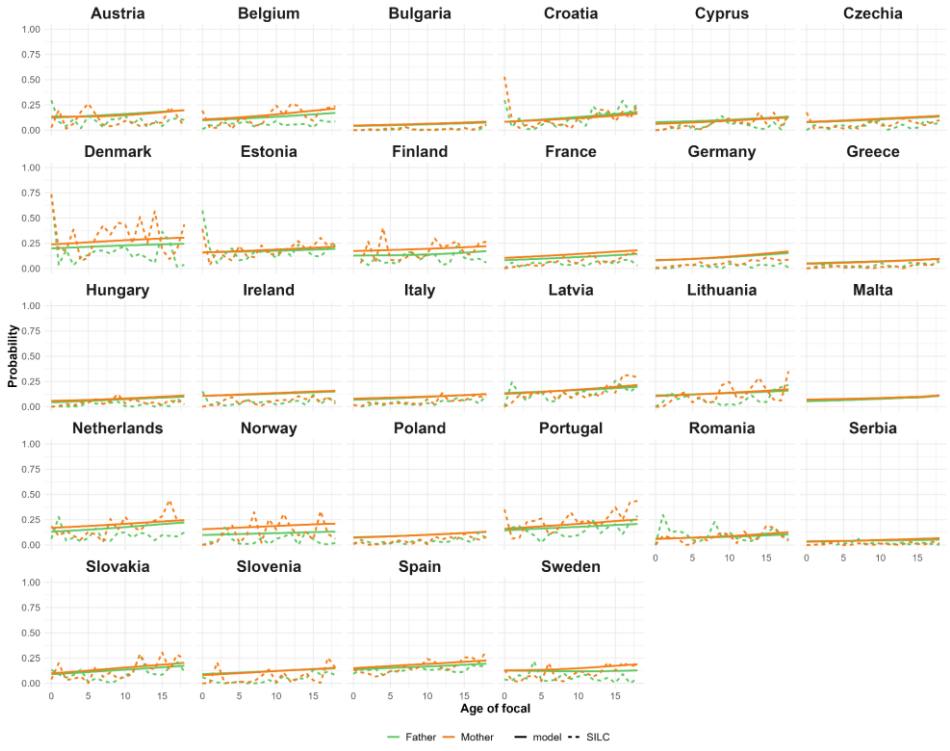


Figure S11: Probability of having at least a parent with disability - comparison between the kinship matrix models and "empirical" ones from SILC data (based on co-resident kin)



Figure S12: Probability of having at least a child with disability - comparison between the kinship matrix models and "empirical" ones from SILC data (based on co-resident kin)

Some general consideration of cross-country differences in exposure to kin disability

The differences observed across countries are likely the result of distinct demographic and health trajectories that European populations have undergone. Fertility and mortality transitions unfolded at different times, with varying pace, intensity, and duration, generating diverse populations, age structures, and kinship configurations across countries (30, 31). Age-specific fertility and survival schedules differ across contexts (see Appendix Figures S13.1 to S13.3), resulting in different pools of kin alive at each age of the focal individual (Appendix Figure S6.1). For example, countries where low exposure is found, such as Norway, which are characterised by comparatively higher and earlier fertility levels in recent decades, tend to exhibit slightly larger and younger numbers of siblings,

children, and grandchildren at most ages compared to high and mid exposure countries, such as Portugal and Italy. These structural differences in kin availability directly shape the size and composition by age and kin type of kinship networks. At the same time, cross-national variation in health and epidemiological trajectories (32) also plays an important role in determining the exposure to kin disability. Age-specific disability prevalence profiles differ across countries (see Appendix, Figure 14.1 and 14.2). For instance, Portugal has consistently higher disability prevalence at most ages than countries with low or medium exposure. This implies that even if two countries had identical kinship structures, the one with higher disability prevalence would exhibit higher proportions of kin with disability and a greater probability of having at least one kin with disability than a lower exposure one (see the differences in the overall number of kin with disability for selected countries in Appendix Figure 6.2). These dynamics can be better captured in future studies aimed at exploring drivers of changes in exposure to kin disability, disentangling how much is driven by each factor.

Data sources

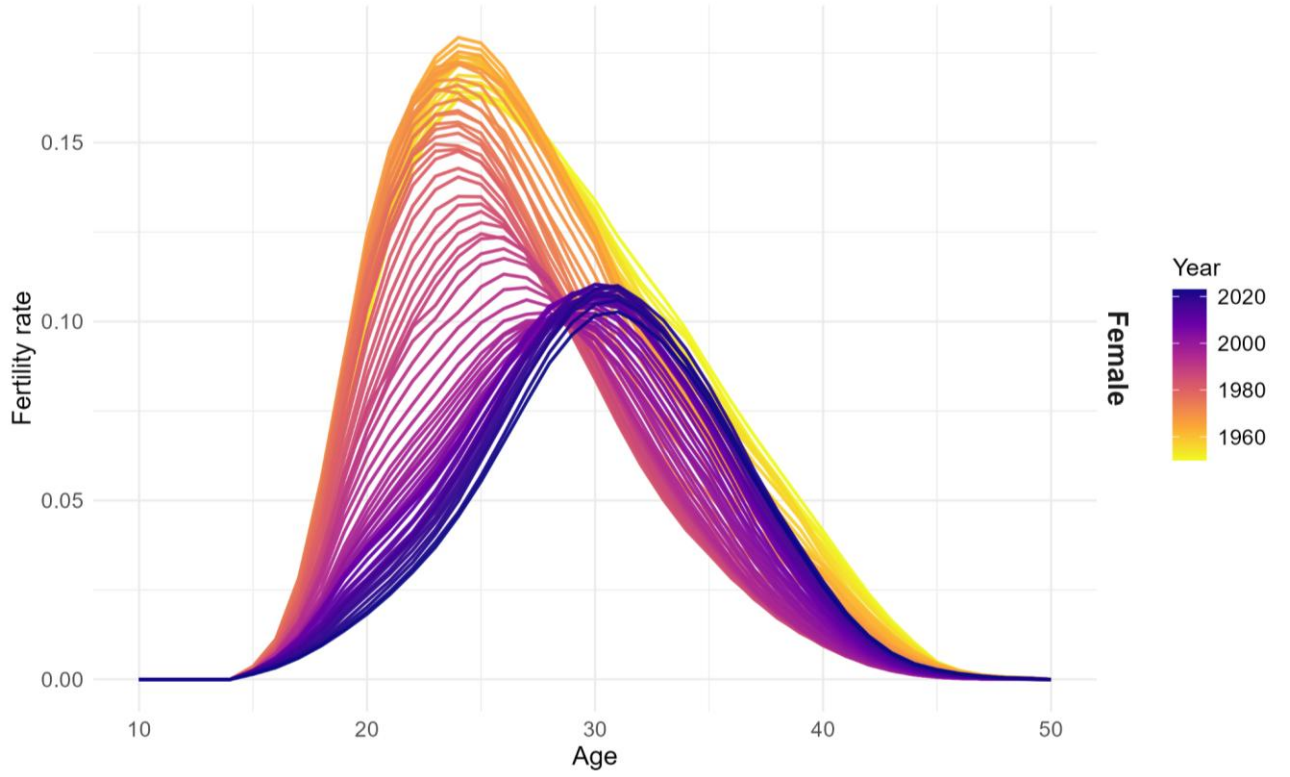


Figure S13.1: Fertility rates by age over time (1950-2023) shown for EU_27. Source: World Population Prospects (59)

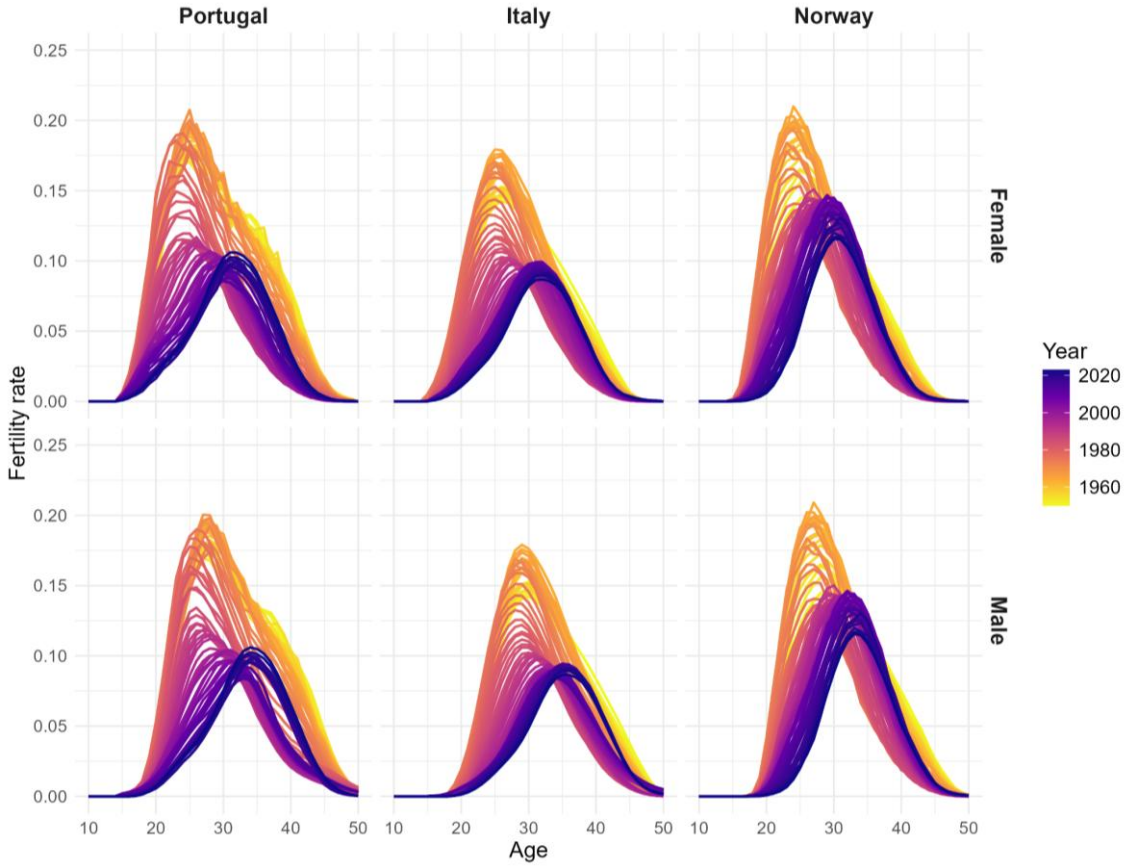
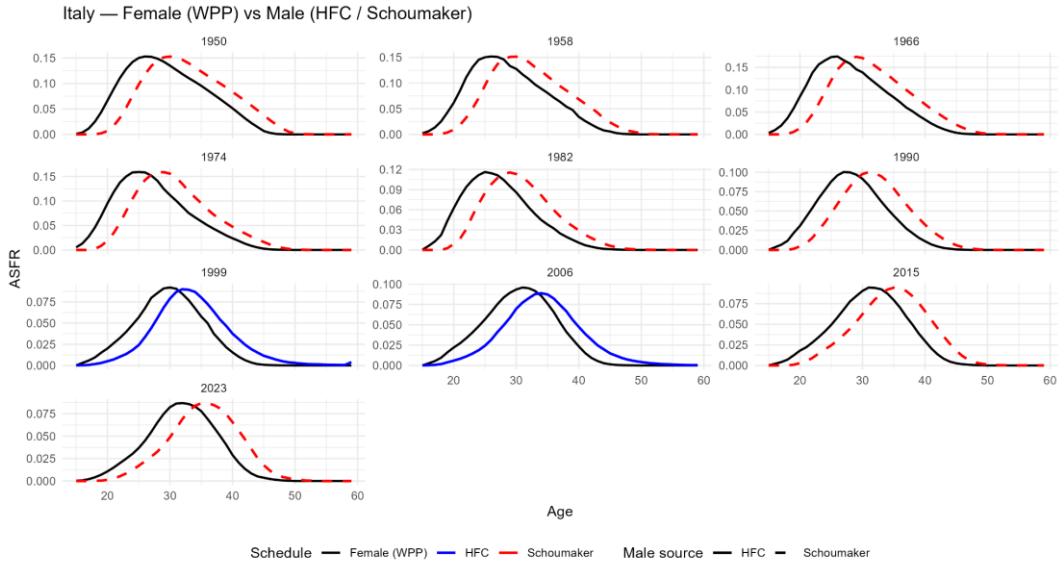


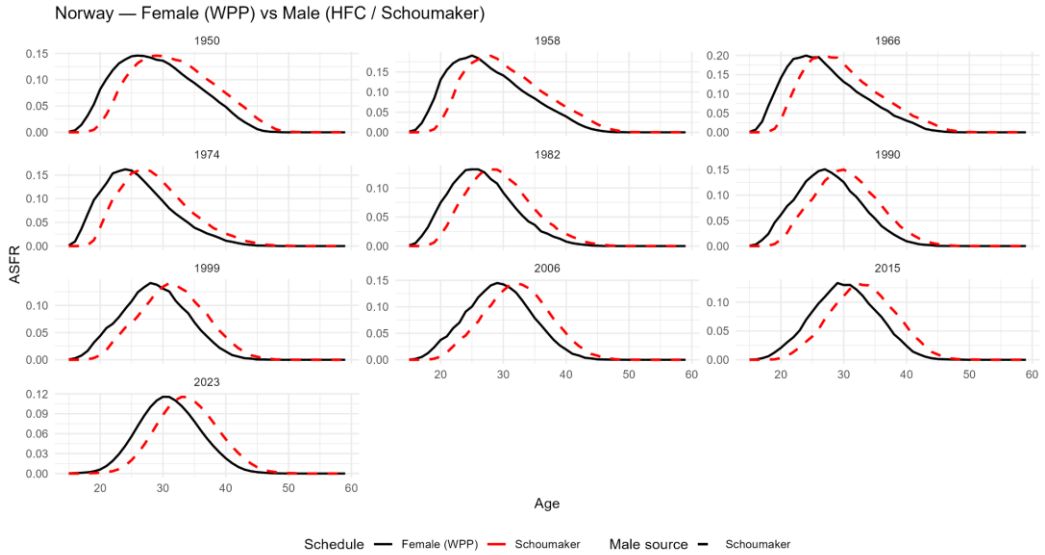
Figure S13.1b: Fertility rates by age over time (1950-2023) shown for selected countries. Source: World Population Prospects (59), HFC, Schoumaker (2019).

Table S3: Sources of male age-specific fertility rates by country and period

Country	Fertility data source and period of coverage
Austria	Schoumaker shift: 1950–2023
Belgium	Schoumaker shift: 1950–2023
Bulgaria	Schoumaker shift: 1950–2023
Croatia	Schoumaker shift: 1950–2023
Cyprus	Schoumaker shift: 1950–2023
Czechia	Schoumaker shift: 1950–2023
Denmark	Schoumaker shift: 1950–1985; HFC: 1986–2015; Schoumaker shift: 2016–2023

Estonia	Schoumaker shift: 1950–1988; HFC: 1989–2014; Schoumaker shift: 2015–2023
Finland	Schoumaker shift: 1950–1986; HFC: 1987–2015; Schoumaker shift: 2016–2023
France	Schoumaker shift: 1950–1997; HFC: 1998–2013; Schoumaker shift: 2014–2023
Germany	Schoumaker shift: 1950–1990; HFC: 1991–2013; Schoumaker shift: 2014–2023
Greece	Schoumaker shift: 1950–2023
Hungary	Schoumaker shift: 1950–1969; HFC: 1970–2014; Schoumaker shift: 2015–2023
Ireland	Schoumaker shift: 1950–2023
Italy	Schoumaker shift: 1950–1998; HFC: 1999–2014; Schoumaker shift: 2015–2023
Latvia	Schoumaker shift: 1950–2023
Lithuania	Schoumaker shift: 1950–2023
Malta	Schoumaker shift: 1950–2023
Netherlands	Schoumaker shift: 1950–2023
Norway	Schoumaker shift: 1950–2023
Poland	Schoumaker shift: 1950–1985; HFC: 1986–2014; Schoumaker shift: 2015–2023
Portugal	Schoumaker shift: 1950–1979; HFC: 1980–2015; Schoumaker shift: 2016–2023
Romania	Schoumaker shift: 1950–2023
Serbia	Schoumaker shift: 1950–2023
Slovakia	Schoumaker shift: 1950–2023
Slovenia	Schoumaker shift: 1950–2023
Spain	Schoumaker shift: 1950–1974; HFC: 1975–2014; Schoumaker shift: 2015–2023
Sweden	Schoumaker shift: 1950–1967; HFC: 1968–2015; Schoumaker shift: 2016–2023


Italy — Female (WPP) vs Male (HFC / Schoumaker)
1950
1958
1966
1974
1982
1990
1999
2006
2015
2023
ASFR
Age
Schedule Female (WPP) HFC Schoumaker
Male source HFC Schoumaker


Norway — Female (WPP) vs Male (HFC / Schoumaker)
1950
1958
1966
1974
1982
1990
1999
2006
2015
2023
ASFR
Age
Schedule Female (WPP) Schoumaker
Male source Schoumaker

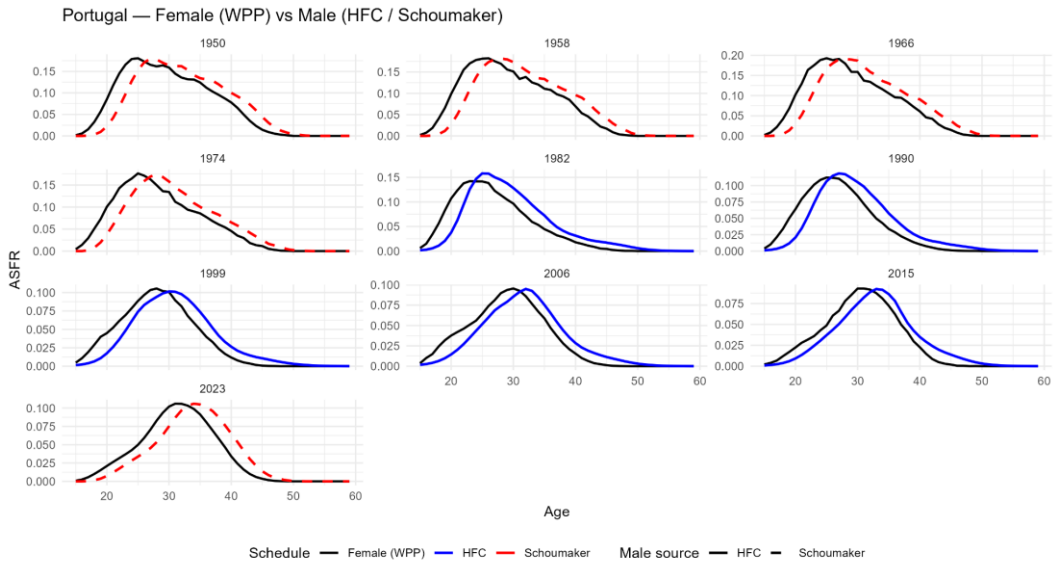

Portugal — Female (WPP) vs Male (HFC / Schoumaker)
1950
1958
1966
1974
1982
1990
1999
2006
2015
2023
ASFR
Age
Schedule Female (WPP) HFC Schoumaker
Male source HFC Schoumaker

Figure S13.2: Male fertility rate comparison, for selected countries, from the different sources used: male fertility = female fertility from World Population Prospects (59), male fertility from HFC (when available), male fertility = female fertility but shifted based on (61) (see methods section)

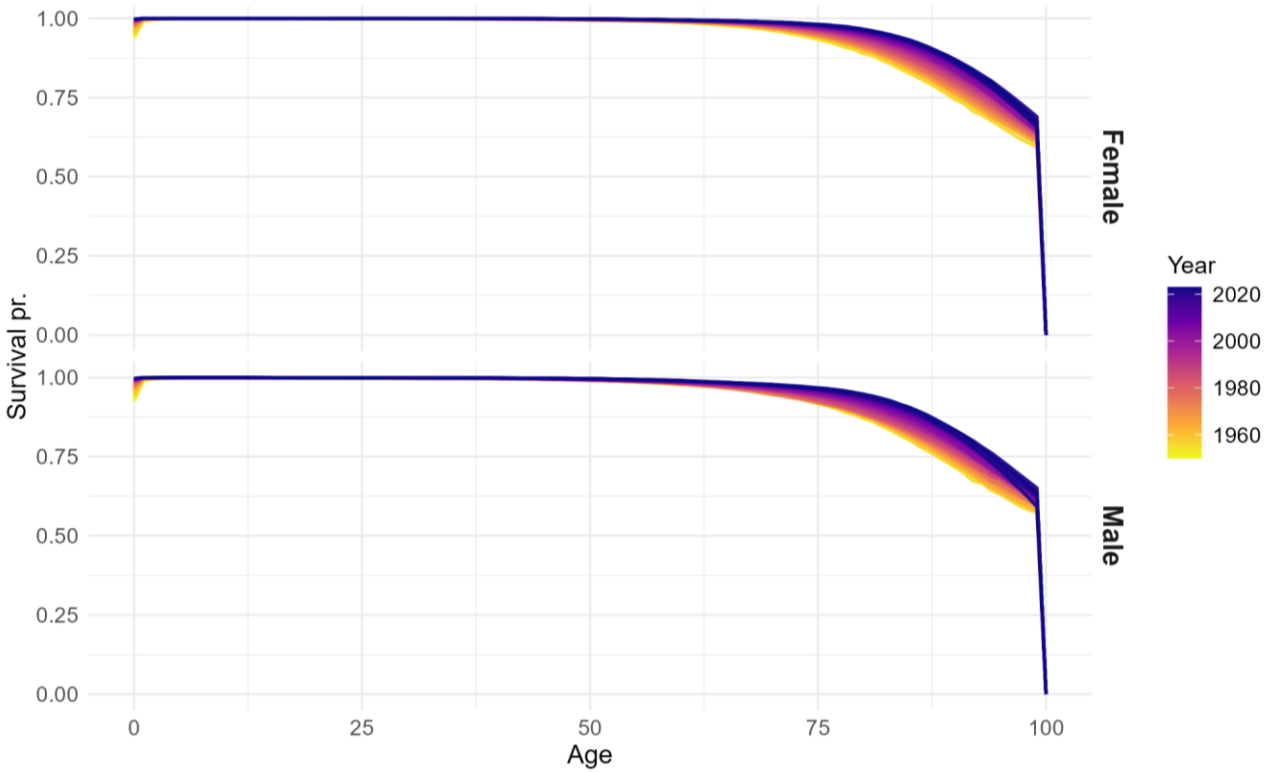


Figure S13.3a Survival probability by age over time (1950-2023) shown for EU_27. Survival probability is set at 0 from age 100. Source: World Population Prospects (59).

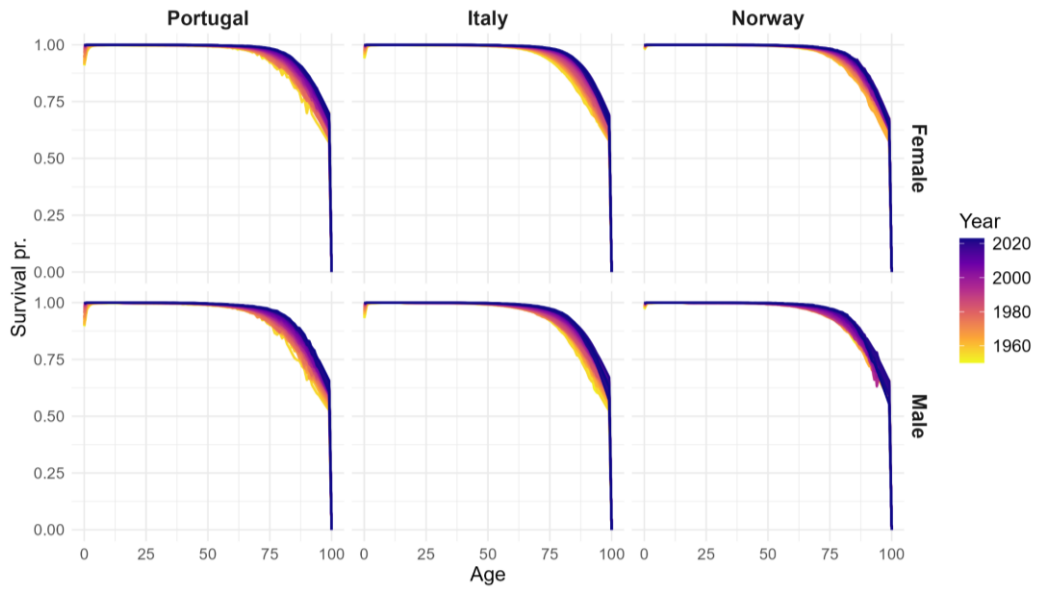


Figure S13.3b Survival probability by age over time (1950-2023) shown for selected countries. Survival probability is set at 0 from age 100. Source: World Population Prospects (59).

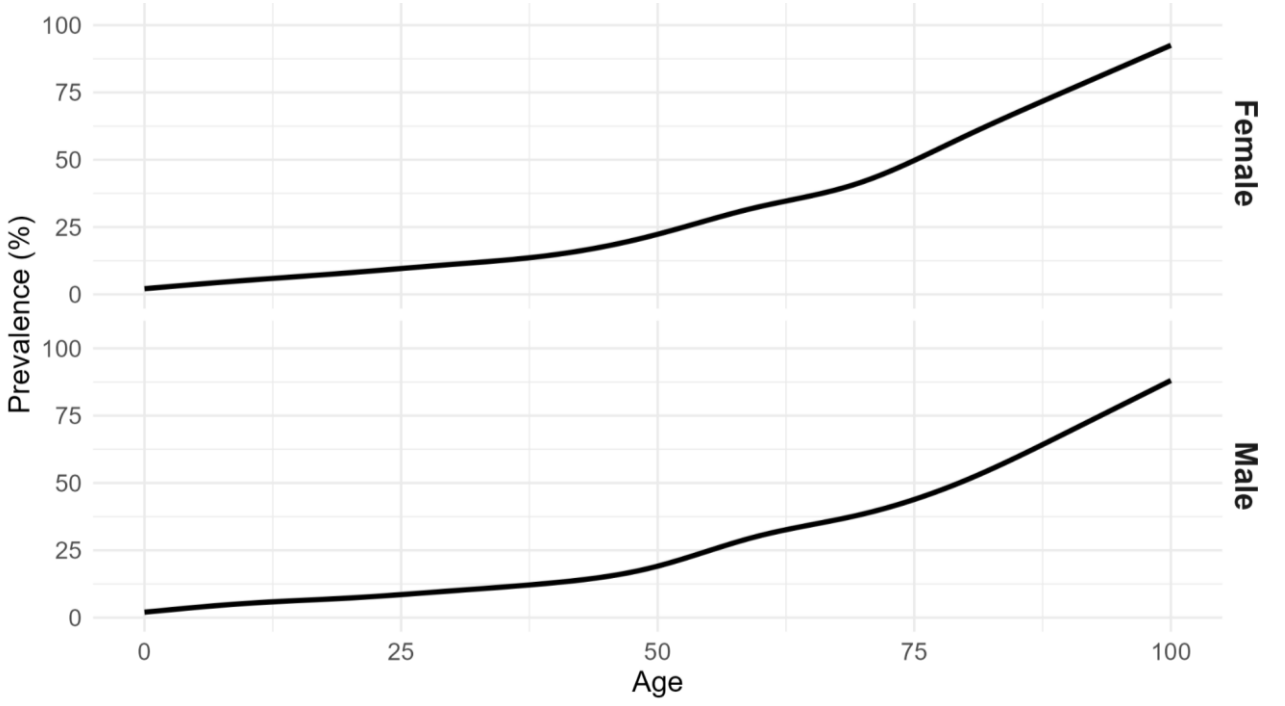


Figure S14.1a Smoothed disability prevalence by gender and age in 2021 for EU_27. Source: own elaboration based on EU-SILC data; non-smoothed prevalence by age classes is available at: https://doi.org/10.2908/HLTH_SILC_12; https://doi.org/10.2908/ILC_HCH13

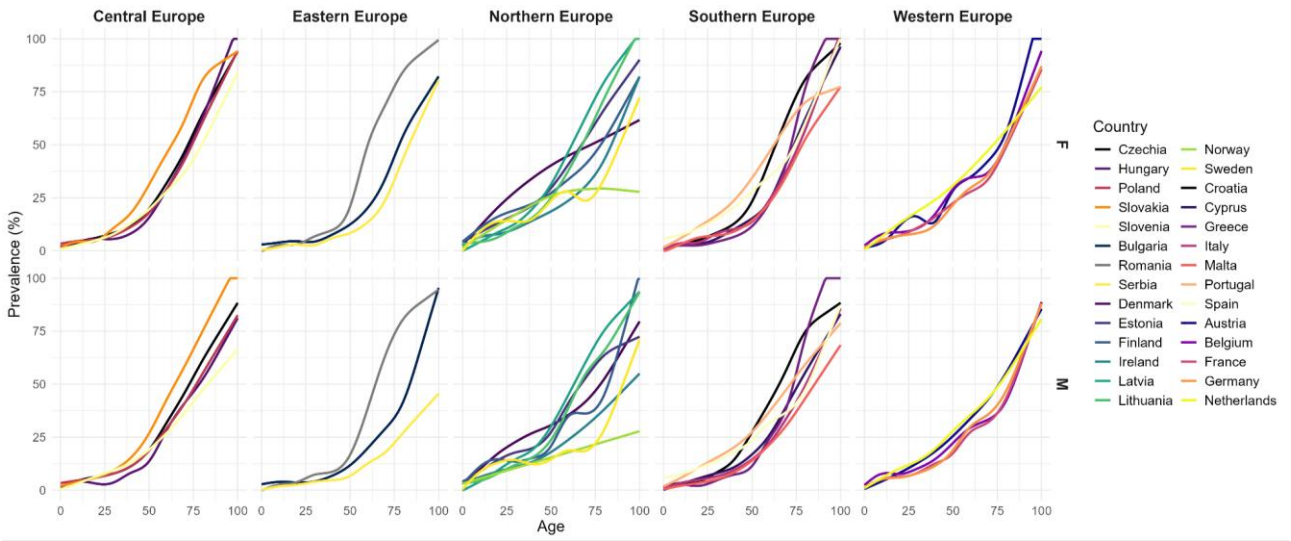


Figure S14.1b Smoothed disability prevalence by gender and age in 2021 for EU countries. Source: own elaboration based on EU-SILC data; non-smoothed prevalence by age classes is available at: https://doi.org/10.2908/HLTH_SILC_12; https://doi.org/10.2908/ILC_HCH13

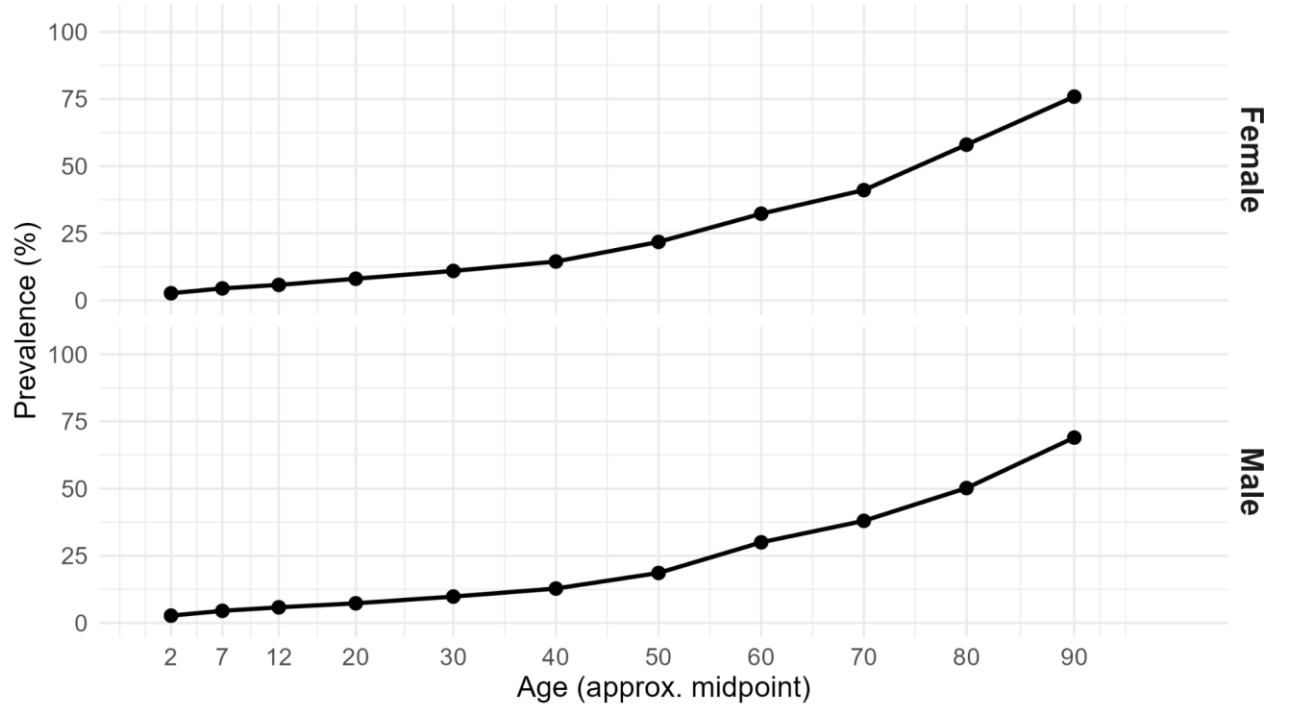


Figure S14.2a Non-smoothed disability prevalence by gender and age in 2021 for EU_27. Source: EU-SILC, available at: https://doi.org/10.2908/HLTH_SILC_12; https://doi.org/10.2908/ILC_HCH13

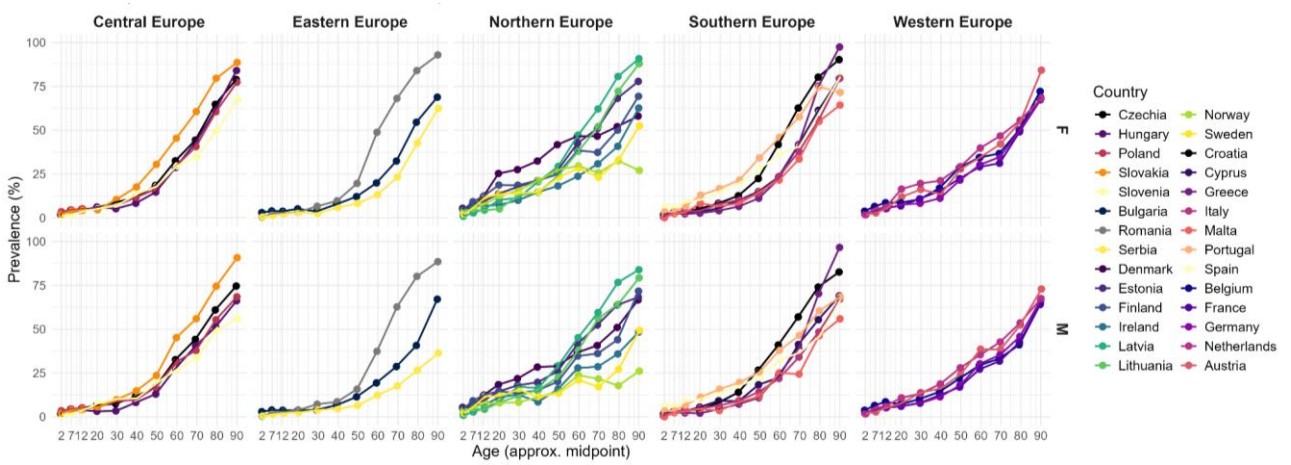


Figure S14.2b Non-smoothed disability prevalence by gender and age in 2021 for EU countries. Source: EU-SILC, available at: https://doi.org/10.2908/HLTH_SILC_12; https://doi.org/10.2908/ILC_HCH13

Survey research disclosure: EU-SILC - GALI

This study uses survey-based disability prevalence estimates from EU-SILC 2021, based on the Global Activity Limitation Indicator (GALI). The authors did not design or administer EU-SILC, recruit respondents, or determine the order of survey questions. EU-SILC data collection, sampling, weighting, non-response adjustment, and fieldwork procedures are carried out by national statistical authorities and harmonised by Eurostat.

The GALI item asks whether respondents have been limited, for at least the past six months, in activities people usually do because of a health problem. Response categories distinguish between "severely limited," "limited but not severely," and "not limited." In this study, individuals reporting either severe or moderate limitations are classified as having disability, while those reporting no limitations are classified as disability-free. For children aged 0–15, the 2021 EU-SILC children's health module provides an analogous measure of activity limitation, which is used to derive childhood disability prevalence.

The main analyses rely on publicly available age- and gender-specific aggregate EU-SILC GALI prevalence estimates for 2021. Individual-level EU-SILC scientific-use files are used only in robustness checks comparing model-based estimates with co-resident kin observed in the survey. Potential effects of sampling design, non-response, and attrition are addressed through the survey procedures and weights provided by Eurostat and national statistical authorities.